Artificial Intelligence, Misinformation, and Market Misconduct

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Artificial intelligence (AI) poses a clear and present danger to our money and our markets. With AI, bad actors and rogue nations can readily and cheaply engage in market manipulation, financial misinformation, and regulatory misconduct that threaten the stability, integrity, and security of our financial system like never before. The intelligence behind this new technology may be artificial, but the damage and losses in market value and confidence are very real.

This Article provides an early, original study of this unsettling reality for our markets, businesses, and society. It begins by surveying the new financial marketplace with the growing adoption of AI and reveals emerging AI-related risks of market misinformation, manipulation, and misconduct. It also examines the pernicious misinformation problem of financial deepfakes. Next, it exposes the larger structural vulnerabilities in the architecture of our technology-driven marketplace. It investigates how the proliferation of autonomous AI technologies creates new, intertwined, and systemic risks related to speed and opacity; exacerbates ongoing geopolitical threats; and increases the likelihood of major financial accidents. The Article then explains how foundational issues relating to regulation, enforcement, resources, and politics limit the capabilities of policymakers and regulators to timely identify and address the dangers posed by AI in finance. Finally, moving from problems to solutions, this Article offers workable recommendations to implement while larger political and policymaking debates unfold on AI and finance. Specifically, it argues for enhancing incentives and penalties for AI-related enforcement actions, encouraging long-term passive retail investments, and revitalizing traditional tools of financial regulation to confront the new challenges posed by AI. Ultimately, this Article aspires to serve as an optimistic, thoughtful, and practical blueprint for better navigating the emerging crossroads of markets, misconduct, misinformation, and AI.

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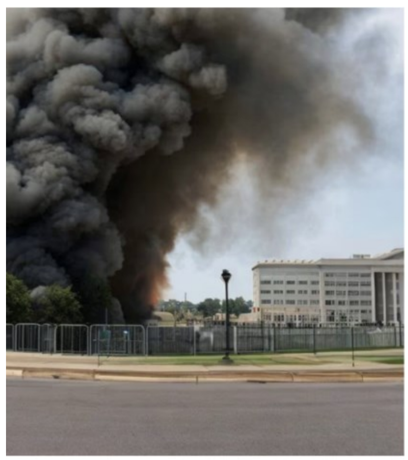
I. INTRODUCTION

One of the next two sentences you read will be a lie. On the morning of Monday, May 22, 2023, around 10:04 a.m., there was an explosion near the Pentagon, the headquarters of the U.S. Department of Defense in Washington, D.C.[1] Reports and pictures of the explosion showing a large dark cloud of smoke went viral on social media, causing both the Dow Jones Industrial Average and the S&P 500 Index, two of the world's leading stock market indicators, to fall significantly between 10:04 a.m. and 10:10 a.m. before fully recovering around 10:13 a.m.[2]

Those images of the explosion, like the one below, turned out to be deepfakes generated by artificial intelligence (AI).[3]

Figure 1: *AI-Generated Deepfake Image of an Explosion at the Pentagon*[4]



It took the markets about nine minutes to fully discern the inauthenticity of the AI-generated fake photos, leading to temporary losses of $500 billion in

[1] Philip Marcelo, *Fact Focus: Fake Image of Pentagon Explosion Briefly Sends Jitters Through Stock Market*, AP (May 23, 2023), https://apnews.com/article/pentagon-explosion-misinformation-stock-market-ai-96f534c790872fde67012ee81b5ed6a4 [https://perma.cc/D4C7-JRCQ]; Donie O'Sullivan & Jon Passantino, *'Verified' Twitter Accounts Share Fake Image of 'Explosion' Near Pentagon, Causing Confusion*, CNN BUS. (May 23, 2023), https://www.cnn.com/2023/05/22/tech/twitter-fake-image-pentagon-explosion/index.html [https://perma.cc/GS66-DQME]; Will Oremus, Drew Harwell & Teo Armus, *A Tweet About a Pentagon Explosion Was Fake. It Still Went Viral*, WASH. POST (May 22, 2023), https://www.washingtonpost.com/technology/2023/05/22/pentagon-explosion-ai-image-hoax.

[2] *See* sources cited *supra* note 1.

[3] Shannon Bond, *Fake Viral Images of an Explosion at the Pentagon Were Probably Created by AI*, NPR (May 22, 2023), https://www.npr.org/2023/05/22/1177590231/fake-viral-images-of-an-explosion-at-the-pentagon-were-probably-created-by-ai [https://perma.cc/2N5Z-E9ZC].

[4] Devesh Kumar, *Fake Image of Explosion at Pentagon Goes Viral, Creates Chaos in Stock Market*, MINT (May 23, 2023), https://www.livemint.com/news/world/aigenerated-image-of-explosion-at-pentagon-goes-viral-creates-chaos-in-stock-market-see-here-11684774645223.html [https://perma.cc/6QE7-Y2KC].

market value.[5] In that brief period of time, those fake images and false reports—amplified by numerous accounts on social media, including ones controlled by the Kremlin and Russian intelligence agencies—spread across the globe before they were officially debunked by the U.S. Department of Defense and the Arlington County Fire Department in Virginia.[6] The intelligence behind those fake images was artificial, but its effects were very real, and so were the billions of dollars in lost market value for investors.[7]

This is the new precarious reality and future of our markets and society. While it will be difficult to precisely predict the long-term impact of AI technologies, two things will likely be true going forward: (1) everything that can be manipulated by AI will be manipulated by AI, and (2) AI will continue to advance and become more sophisticated in ways that we cannot foresee.[8] And so, today's challenges and dangers are truly precursors to the more daunting challenges and dangers posed by tomorrow's AI-driven market manipulation.

This Article is about this new ominous reality of our financial markets and its consequences for our businesses, markets, and money in the face of rapidly rising risks posed by AI.[9] While there are many ongoing debates and discussions about various hypothetical, abstract, and future dangers of AI, this Article is about a very clear and present danger that requires closer examination and urgent action.[10] To that end, this Article offers an early, original study of the

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[5] Ian Krietzberg, *S&P Sheds $500 Billion from Fake Pentagon Explosion*, THESTREET (May 22, 2023), https://www.thestreet.com/technology/s-p-sheds-500-billion-from-fake-pentagon-explosion [https://perma.cc/UUF5-WYTK]; Marcelo, *supra* note 1; O'Sullivan & Passantino, *supra* note 1.

[6] *See* Andrew Ross Sorkin et al., *An A.I.-Generated Spoof Rattles the Markets*, N.Y. TIMES (May 23, 2023), https://www.nytimes.com/2023/05/23/business/ai-picture-stock-market.html; Kierra Frazier, *Fake Image of Pentagon Explosion Goes Viral*, POLITICO (May 22, 2023), https://www.politico.com/news/2023/05/22/fake-image-pentagon-explosion-goes-viral-00098207; Bond, *supra* note 3.

[7] Krietzberg, *supra* note 5.

[8] *See, e.g.*, ETHAN MOLLICK, CO-INTELLIGENCE: LIVING AND WORKING WITH AI 58–62 (2024) (discussing continuing advances in AI).

[9] *See* William Magnuson, *Regulating Fintech*, 71 VAND. L. REV. 1167, 1168–70 (2018) (discussing the rise of financial technologies like those powered by AI); Thomas H. Davenport & Rajeev Ronanki, *Artificial Intelligence for the Real World*, HARV. BUS. REV., Jan.–Feb. 2018, at 108, 108–16 (discussing various applications of AI, their benefits, their challenges, and the future of AI). *See generally* ARTIFICIAL INTELLIGENCE IN FINANCIAL MARKETS (Christian L. Dunis et al. eds., 2016) (discussing how AI is currently being used in finance).

[10] *See* Dan Milmo, *Hope or Horror? The Great AI Debate Dividing Its Pioneers*, GUARDIAN (Oct. 24, 2023), https://www.theguardian.com/technology/2023/oct/24/hope-or-horror-the-great-ai-debate-dividing-its-pioneers [https://perma.cc/NX8C-JMS8]; Cade Metz & Gregory Schmidt, *Elon Musk and Others Call for Pause on A.I., Citing 'Profound Risks to Society*,*'* N.Y. TIMES (Mar. 29, 2023), https://www.nytimes.com/2023/03/29/technology/ai-artificial-intelligence-musk-risks.html; Kevin Roose, *Inside the White-Hot Center of A.I. Doomerism*, N.Y. TIMES (July 11, 2023), https://www.nytimes.com/2023/07/11/technology/

growing threats posed by AI-driven financial misconduct and misinformation, reveals systemic vulnerabilities in this new informational marketplace, identifies regulatory challenges to governing AI in finance, and recommends workable next steps to better safeguard the stability of our economy and the integrity of our financial markets.[11]

Drawing and building on the author's prior research as well as a rich and growing body of interdisciplinary literature on financial regulation and AI that spans law, finance, management, sociology, and technology, this Article seeks to make three contributions.[12] First, this Article aims to provide a preliminary account for understanding, explaining, and debating AI-powered manipulation, misconduct, and misinformation in our financial markets. Second, it aims to identify and analyze larger systemic and structural challenges associated with regulating and governing issues relating to AI in modern financial markets.

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anthropic-ai-claude-chatbot.html; Cade Metz, *How Could A.I. Destroy Humanity?*, N.Y. TIMES (June 10, 2023), https://www.nytimes.com/2023/06/10/technology/ai-humanity.html.

[11] AI, as a concept and technology, is difficult to precisely define because of its dynamic nature and rapidly evolving applications. For the purpose of this Article, AI herein refers to a variety of technology and tools that can perform various tasks autonomously or semi-autonomously in a manner that is akin to actions resulting from human intelligence. *See, e.g.*, Cynthia Dwork & Martha Minow, *Distrust of Artificial Intelligence: Sources & Responses from Computer Science & Law*, DÆDALUS, Spring 2022, at 309, 310 ("Artificial intelligence signifies a variety of technologies and tools that can solve tasks requiring 'perception, cognition, planning, learning, communication, and physical actions,' often learning and acting without oversight by their human creators or other people."); 15 U.S.C. § 9401(3) ("The term 'artificial intelligence' means a machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations or decisions influencing real or virtual environments.").

[12] *See, e.g.*, HILARY J. ALLEN, DRIVERLESS FINANCE: FINTECH'S IMPACT ON FINANCIAL STABILITY 47–51 (2022); STUART RUSSELL, HUMAN COMPATIBLE: ARTIFICIAL INTELLIGENCE AND THE PROBLEM OF CONTROL 103–10 (2019); AJAY AGRAWAL, JOSHUA GANS & AVI GOLDFARB, PREDICTION MACHINES: THE SIMPLE ECONOMICS OF ARTIFICIAL INTELLIGENCE 24–30 (2018); FRANK PASQUALE, THE BLACK BOX SOCIETY: THE SECRET ALGORITHMS THAT CONTROL MONEY AND INFORMATION 186–87 (2015); NICK BOSTROM, SUPERINTELLIGENCE: PATHS, DANGERS, STRATEGIES 127–30 (2014); JERRY W. MARKHAM, LAW ENFORCEMENT AND THE HISTORY OF FINANCIAL MARKET MANIPULATION 375 (2014); SCOTT PATTERSON, DARK POOLS: THE RISE OF THE MACHINE TRADERS AND THE RIGGING OF THE U.S. STOCK MARKET 322–23 (2013); BRIAN R. BROWN, CHASING THE SAME SIGNALS: HOW BLACK-BOX TRADING INFLUENCES STOCK MARKETS FROM WALL STREET TO SHANGHAI 113 (2010); Margot E. Kaminski, *Regulating the Risks of AI*, 103 B.U. L. REV. 1347, 1350 (2023); Jill E. Fisch, *GameStop and the Reemergence of the Retail Investor*, 102 B.U. L. REV. 1799, 1802–03 (2022); Kristen E. Eichensehr, *The Law and Politics of Cyberattack Attribution*, 67 UCLA L. REV. 520, 532 n.39 (2020); Dan Awrey & Kathryn Judge, *Why Financial Regulation Keeps Falling Short*, 61 B.C. L. REV. 2295, 2306–07 (2020); Bobby Chesney & Danielle Citron, *Deep Fakes: A Looming Challenge for Privacy, Democracy, and National Security*, 107 CALIF. L. REV. 1753, 1758 (2019); Andrei A. Kirilenko & Andrew W. Lo, *Moore's Law Versus Murphy's Law: Algorithmic Trading and Its Discontents*, J. ECON. PERSPS., Spring 2013, at 51, 52.

Third, it aims to recommend a set of pragmatic principles and proposals for public policymakers and business stakeholders to better navigate and protect our marketplace and investments against the risks that arise at the intersection of AI and finance.

In seeking these objectives, this Article is mindful of the fact that rapidly evolving AI technologies and their use cases will necessarily result in this paper being dated upon publication. Chances are good that by the time you are done reading this sentence, AI technologies and their applications will have advanced in incremental but meaningful ways. In fact, the highly dynamic, evolving nature of AI has made it difficult to even define it as a concept or technology for purposes of law and regulation.[13] Nevertheless, however dated, such a pursuit is also a worthy one, for it can reveal the deep and consequential unfolding changes in our marketplace and lay the principled groundwork for future paths of financial regulation, corporate governance, and technological innovation. Ultimately, this Article endeavors to provide a draft blueprint for regulators, executives, and investors to better conceptualize and respond to critical issues at the crossroads of AI, markets, misconduct, and misinformation.

This Article constructs this blueprint in four parts. Part II surveys the new financial marketplace with the growing adoption of AI. It critically examines how earlier forms of AI changed financial market operations and how more recent forms of AI pose a different, ominous existential shift to the marketplace while bringing out many potential lasting benefits. In particular, it explains how and why the integration of new information technologies over the past quarter-century sowed the seeds of emerging AI-related risks of market misinformation, manipulation, and misconduct. It also explores the pernicious misinformation problem of financial deepfakes in the marketplace and assesses the high long-term costs of AI-fueled financial misconduct in terms of trust erosion.

Part III highlights larger structural vulnerabilities in the architecture of the AI-driven modern financial markets. It investigates how proliferating informational and operational AI tools create new, intertwined systemic risks, exacerbates ongoing geopolitical threats, and increases the odds of major accidents for the financial system and the greater economy.[14] Specifically, Part

[13] *See, e.g.*, Bryan Casey & Mark A. Lemley, *You Might Be a Robot*, 105 CORNELL L. REV. 287, 287 (2020) ("No one has been able to offer a decent definition of robots and AI—not even experts."); Kaminski, *supra* note 12, at 1355; Matt O'Shaughnessy, *One of the Biggest Problems in Regulating AI Is Agreeing on a Definition*, CARNEGIE ENDOWMENT FOR INT'L PEACE (Oct. 6, 2022), https://carnegieendowment.org/2022/10/06/one-of-biggest-problems-in-regulating-ai-is-agreeing-on-definition-pub-88100 [https://perma.cc/77TN-V8HH] ("As policymakers around the world have attempted to create guidance and regulation for AI's use in settings ranging from school admissions and home loan approvals to military weapon targeting systems, they all face the same problem: AI is *really* challenging to define.").

[14] *See, e.g.*, BRAD SMITH & CAROL ANN BROWNE, TOOLS AND WEAPONS: THE PROMISE AND THE PERIL OF THE DIGITAL AGE 69–76 (2019) (discussing how technological tools can

III begins by exposing how AI-related misinformation can generate new systemic risks involving speed and opacity. Next, it highlights how AI can compound existing geopolitical tensions by empowering adversaries with inexpensive weapons to attack and disrupt our financial system given the "dual use" nature of AI technologies for civilian and military objectives.[15] Finally, it reveals how the technologically networked complexities of AI systems increase the odds of significant glitches and accidents in our financial markets.

Shifting from endogenous risks to exogenous obstacles, Part IV examines regulatory and policy challenges for addressing the dangers related to using and misusing AI in our financial markets. It explores how foundational issues relating to regulation, enforcement, and resources limit the capabilities of regulators to properly identify and address the risks posed by AI in finance. Additionally, it reveals how domestic and international politics hinder the ability of regulators to meaningfully confront the dangers posed by AI in finance as regulators attempt to craft new rules and paths for the road ahead.[16]

Moving beyond sketching problems and challenges, Part V offers workable recommendations for the near future while the larger, long-term political and policymaking debates unfold on AI and finance. It starts by recommending enhanced enforcement incentives and penalties from regulators like the Securities and Exchange Commission (SEC) and Department of Justice (DOJ) to bolster private capabilities for guarding against the financial risks and threats posed by AI-powered misconduct and misinformation. Next, it argues for policies and practices that encourage long-term investment strategies for investors, particularly retail investors, to better safeguard against the volatility of an AI-powered financial marketplace. Part V closes with a call to revitalize traditional regulatory tools like public disclosures, human examinations, and stress tests to better mitigate the financial risks associated with AI, misconduct, and misinformation.

Finally, this Article ends with a brief conclusion that reflects on the unfolding and evolving regulatory architecture of our financial markets in the emerging era of AI. It looks forward with hope that more creative and practical perspectives can help generate a more prosperous society and secure economy for all of us.

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be used as tools for human enrichment as well as weapons for serious social and economic harm).

[15] NAT'L SEC. COMM'N ON A.I., FINAL REPORT 22 (2021), https://assets.foleon.com/eu-central-1/de-uploads-7e3kk3/48187/nscai_full_report_digital.04d6b124173c.pdf ("AI is the quintessential 'dual use' technology—it can be used for civilian and military purposes.").

[16] *See, e.g.*, Adam Satariano & Cecilia Kang, *How Nations Are Losing a Global Race to Tackle A.I.'s Harms*, N.Y. TIMES (Dec. 6, 2023), https://www.nytimes.com/2023/12/06/technology/ai-regulation-policies.html.

II. AI, MARKETS, AND MISCONDUCT

Over the past half-century, the financial industry has increasingly harnessed new technologies like AI to improve and revolutionize various aspects of its operations and services.[17] Financial technologies have improved liquidity, enhanced efficiencies, expanded access, and helped grow the size and productivity of the marketplace.[18] Early forms of financial technology and AI brought with them various net benefits for financial markets, but the more recent advanced forms present serious, unprecedented risks to the stability and integrity of the marketplace through AI-powered misinformation and market misconduct.[19] If left unchecked and unregulated, these risks can result in less trust in the marketplace, higher costs of capital, and fewer of the benefits that AI can bring to society.

A. *AI and Financial Markets*

Even before the wave of attention and innovation brought forth by the public launch of OpenAI's ChatGPT in 2023, the financial industry had long been an early and enthusiastic adopter of AI and new technologies.[20] Advances in computing, semiconductors, and information technology over the last few

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[17] *See, e.g.*, Magnuson, *supra* note 9, at 1169 ("[T]he fintech revolution promises to produce great benefits for the wider economy, including broader access to capital, fairer lending standards, better investment advice, and more secure transactions.").

[18] *See* Kirilenko & Lo, *supra* note 12, at 52 ("Algorithmic trading is part of a much broader trend in which computer-based automation has improved efficiency by lowering costs, reducing human error, and increasing productivity.").

[19] *See, e.g.*, ALLEN, *supra* note 12, at 3 ("Fintech innovation does have the potential to deliver financial services more quickly and cost effectively, particularly to people who haven't been able to access mainstream financial services before."); Douglas W. Arner, Jànos Barberis & Ross P. Buckley, *FinTech, RegTech, and the Reconceptualization of Financial Regulation*, 37 NW. J. INT'L L. & BUS. 371, 377–81 (2017).

[20] *See, e.g.*, Vauhini Vara, *One Year In and ChatGPT Already Has Us Doing Its Bidding*, N.Y. TIMES (Dec. 19, 2023), https://www.nytimes.com/2023/12/19/opinion/artificial-intelligence-chatgpt.html; David Streitfeld, *Silicon Valley Confronts the Idea that the 'Singularity' Is Here*, N.Y. TIMES (June 11, 2023), https://www.nytimes.com/2023/06/11/technology/silicon-valley-confronts-the-idea-that-the-singularity-is-here.html; Farhad Manjoo, *ChatGPT Is Already Changing How I Do My Job*, N.Y. TIMES (Apr. 21, 2023), https://www.nytimes.com/2023/04/21/opinion/chatgpt-journalism.html; DAVID J. LEINWEBER, NERDS ON WALL STREET: MATH, MACHINES, AND WIRED MARKETS 31–63 (2009) (chronicling the computerization of financial markets); Jonathan R. Macey & Maureen O'Hara, *From Markets to Venues: Securities Regulation in an Evolving World*, 58 STAN. L. REV. 563, 563 (2005) ("Advances in technology, combined with the dramatic decrease in the cost of information processing, have conspired to change the way that securities transactions occur.").

decades have had profound effects on society and business.[21] Finance is no exception; in fact, it is arguably one of the most changed areas of society as a result of advances in technology in the last half-century.[22]

Once human-dominated, financial operations have been supplemented, augmented, or outright eliminated by AI-powered machines.[23] AI and related technologies have fundamentally altered key operations of modern finance.[24] This adoption and integration of AI in finance is propelled in part by the industry's intrinsic dependence on processing vast amounts of financial data to generate valuable insights at incredibly high speeds, and one of AI's core competencies is analyzing and extracting valuable intelligence from troves of information in ways that are not humanly possible.[25] Furthermore, once insight is derived from the troves of data, AI-driven financial machines can learn to revise and develop algorithmic codes to respond to new data and scenarios on their own.[26]

Financial institutions have leveraged AI technologies for a wide range of applications in their operations, with advances in technologies leading to a proliferation in use cases.[27] AI-powered algorithms execute trades with

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[21] *See* Kirilenko & Lo, *supra* note 12, at 51 ("Over the past four decades, the remarkable growth of the semiconductor industry as embodied by Moore's Law has had enormous effects on society, influencing everything from household appliances to national defense.").

[22] *See* SUBCOMM. ON EMERGING & EVOLVING TECHS., U.S. COMMODITY FUTURES TRADING COMM'N, RESPONSIBLE ARTIFICIAL INTELLIGENCE IN FINANCIAL MARKETS: OPPORTUNITIES, RISKS & RECOMMENDATIONS 36–37 (2024), https://www.cftc.gov/media/10626/TAC_AIReport050224/download [https://perma.cc/8QRQ-YBF3] (chronicling the growth and evolution of AI technologies in finance).

[23] *See* Tom C.W. Lin, *The New Investor*, 60 UCLA L. REV. 678, 687–93 (2013).

[24] *See, e.g.*, Felix Salmon & Jon Stokes, *Algorithms Take Control of Wall Street*, WIRED (Dec. 27, 2010), https://www.wired.com/2010/12/ff-ai-flashtrading [https://perma.cc/3LY7-MDJU] ("It's the machines' market now; we just trade in it.").

[25] *See* ALLEN, *supra* note 12, at 51 ("Machine learning also differs from human learning in that it is capable of processing far more data, and has the capacity to see relationships between many more variables, than a human could ever manage."); Gina-Gail S. Fletcher & Michelle M. Le, *The Future of AI Accountability in the Financial Markets*, 24 VAND. J. ENT. & TECH. L. 289, 291 (2022) ("The impact of AI is particularly prominent in the financial markets. The ability of algorithms to quickly process vast quantities of data makes them valuable tools for financial institutions where time is money and data is king." (footnote omitted)).

[26] *See* ALLEN, *supra* note 12, at 51 ("[M]achine learning algorithms scour data, detect patterns and then use those patterns to construct their own decision-making rules. Armed with these decision-making rules, algorithms can figure out how to respond to any new data provided to them.").

[27] *See, e.g.*, Concept Release on Risk Controls and System Safeguards for Automated Trading Environments, 78 Fed. Reg. 56542, 56573 app. 2 (Sept. 12, 2013) (to be codified at 17 C.F.R. ch. 1) ("We have witnessed a fundamental shift in markets from human-based trading to highly automated electronic trading."); Bradley Hope, *How Computers Trawl a Sea of Data for Stock Picks*, WALL ST. J. (Apr. 1, 2015), http://www.wsj.com/articles/how-

remarkable results at speeds and volumes beyond human capabilities.[28] Risk-assessment models use machine learning to assess the investment-worthiness of a wide array of assets.[29] AI is assisting research analysts in uncovering new insights on specific instruments, companies, and markets.[30] Compliance systems are using AI to spot and troubleshoot patterns of potential misconduct, fraud, and vulnerability within firms.[31] Furthermore, AI-driven chatbots and virtual assistants have enhanced some areas of customer service, providing more efficient, responsive, and personalized client interactions.[32]

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computers-trawl-a-sea-of-data-for-stock-picks-1427941801; Sheelah Kolhatkar & Sree Vidya Bhaktavatsalam, *The Colossus of Wall Street*, BLOOMBERG (Dec. 9, 2010), https://www.bloomberg.com/news/articles/2010-12-09/the-colossus-of-wall-street; Andrew Ross Sorkin, *Fintech Firms Are Taking on the Big Banks, but Can They Win?*, N.Y. TIMES (Apr. 6, 2016), https://www.nytimes.com/2016/04/07/business/dealbook/fintech-firms-are-taking-on-the-big-banks-but-can-they-win.html; Seth Stevenson, *The Wolf of Wall Tweet*, SLATE (Apr. 20, 2015), http://www.slate.com/articles/business/moneybox/2015/04/bot_makes_2_4_million_reading_twitter_meet_the_guy_it_cost_a_fortune.html//; John F. Wasik, *Sites to Manage Personal Wealth Gaining Ground*, N.Y. TIMES (Feb. 10, 2014), https://www.nytimes.com/2014/02/11/your-money/sites-to-manage-personal-wealth-gaining-ground.html.

28 PATTERSON, *supra* note 12, at 322–23.

29 *See, e.g.*, ALLEN, *supra* note 12, at 51 ("Machine learning is now being incorporated into the financial risk modeling process."); Kolhatkar & Bhaktavatsalam, *supra* note 27; Abbas Basrai & Slim Ben Ali, *Artificial Intelligence in Risk Management*, KPMG, https://kpmg.com/ae/en/home/insights/2021/09/artificial-intelligence-in-risk-management.html [https://perma.cc/LTP4-QXZ8]; Paulette Perhach, *Would You Take Financial Advice from A.I.?*, N.Y. TIMES (May 20, 2023), https://www.nytimes.com/2023/05/20/business/ai-financial-advice-chatgpt.html; William Magnuson, *Artificial Financial Intelligence*, 10 HARV. BUS. L. REV. 337, 339–42 (2020).

30 *See, e.g.*, *Financial Planning and Analysis*, IBM, https://www.ibm.com/products/planning-analytics/financial-planning-analysis [https://perma.cc/4W5U-3ZCZ]; Ronit Ghose, *Artificial Intelligence in Finance – Opportunities & Risks*, CITIGROUP (July 31, 2023), https://www.citigroup.com/global/insights/global-insights/artificial-intelligence-in-finance-opportunities-risks [https://perma.cc/F8B2-WPSJ]; MELISSA KOIDE, BROOKINGS, AI FOR GOOD: RESEARCH INSIGHTS FROM FINANCIAL SERVICES (2022), https://www.brookings.edu/wp-content/uploads/2022/08/AI-for-good-Research-insights-from-financial-services-3.pdf [https://perma.cc/DZY8-5FPP].

31 *See, e.g.*, *Dow Jones Risk & Compliance: Data and Risk Management*, DOW JONES https://www.dowjones.com/professional/risk [https://perma.cc/AH2Z-JSZK]; Don Johnson & Alex Treuber, *How AI Will Affect Compliance Organizations*, ERNST & YOUNG (July 18, 2023), https://www.ey.com/en_us/financial-services/how-ai-will-affect-compliance-organizations [https://perma.cc/94FR-BUFY]; *How Is Financial Services Leveraging AI for Compliance Purposes?*, FINTECH GLOB. (Mar. 30, 2023), https://fintech.global/2023/03/30/how-is-financial-services-leveraging-ai-for-compliance-purposes [https://perma.cc/PXM5-DCEG].

32 *See, e.g.*, Kevin Roose, *Personalized A.I. Agents Are Here. Is the World Ready for Them?* N.Y. TIMES, https://www.nytimes.com/2023/11/10/technology/personalized-ai-agents.html (Nov. 11, 2023); Roman Viliavin, *Customer Support: Using AI Chatbots for Efficiency and Empathy*, FORBES: BUS. DEV. COUNCIL (July 18, 2023), https://www.forbes.com/sites/forbesbusinessdevelopmentcouncil/2023/07/18/customer-support-

Beyond the institutional applications and uses, AI has also changed how individuals manage their money. Software powered by AI helps individuals bank and budget.[33] AI-driven apps help retail investors identify new investments, reallocate current holdings, and learn about new opportunities.[34] Much of this can be done from the convenience of their phones without dealing with a costly intermediary like a traditional broker, financial advisor, or research analyst.[35]

Overall, the financial sector's adoption of the latest AI technologies reflects a broader trend aimed at improving operational efficiency, investor access, and error reduction by leveraging new financial technology over the last two decades.[36] AI, particularly its later generative iteration, is a significant accelerant of these effects due to its capability to harvest, digest, and synthesize enormous amounts of financial data and information and then translate it into actionable intelligence for its end users, be they sophisticated institutions or ordinary investors.[37] And with each passing hour, AI programs using large language models and other underlying technologies continue to advance in a variety of financial and nonfinancial functions.[38]

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using-ai-chatbots-for-efficiency-and-empathy [https://perma.cc/8HL2-77PY]; Brian X. Chen, Nico Grant & Karen Weise, *How Siri, Alexa and Google Assistant Lost the A.I. Race*, N.Y. TIMES (Mar. 15, 2023), https://www.nytimes.com/2023/03/15/technology/siri-alexa-google-assistant-artificial-intelligence.html.

[33] *See, e.g.*, *Your Ultimate AI-Powered Banking Chatbot*, IBM, https://www.ibm.com/products/watsonx-assistant/banking [https://perma.cc/57GW-B2XF]; Alina Tugend, *A Smarter App Is Watching Your Wallet*, N.Y. TIMES, https://www.nytimes.com/2021/03/09/business/apps-personal-finance-budget.html (Mar. 29, 2021).

[34] Wasik, *supra* note 27; Elizabeth MacBride, *Using AI, Investing.com Aims to Be the Bloomberg of Retail Investing*, FORBES, https://www.forbes.com/sites/elizabethmacbride/2023/11/21/using-ai-investingcom-aims-to-be-the-bloomberg-of-retail-investing [https://perma.cc/KX3D-G2U2] (Nov. 21, 2023); Yun Li, *ChatGPT Meets Robinhood? New Investing App Features AI-Powered Portfolio Mentor*, CNBC (Apr. 27, 2023), https://www.cnbc.com/2023/04/27/chatgpt-meets-robinhood-new-app-features-ai-powered-portfolio-mentor-.html [https://perma.cc/CC9N-SYJJ].

[35] *See* Li, *supra* note 34.

[36] *See, e.g.*, Rob Copeland, *The Worst Part of a Wall Street Career May Be Coming to an End*, N.Y. TIMES, https://www.nytimes.com/2024/04/10/business/investment-banking-jobs-artificial-intelligence.html (Apr. 11, 2024) (reporting on how AI may replace junior bankers on many investment banking tasks); ALLEN, *supra* note 12, at 3 (acknowledging that new financial technology can lower costs and expand access, particularly for underserved populations); Joel Hasbrouck & Gideon Saar, *Low-Latency Trading*, 16 J. FIN. MKTS. 646, 648 (2013) (discussing the market-stabilizing attributes of autonomous high-frequency systems); Charles R. Korsmo, *High-Frequency Trading: A Regulatory Strategy*, 48 U. RICH. L. REV. 523, 549–50 (2014) (noting the benefits to investors arising from AI-powered high-frequency trading).

[37] *See, e.g.*, Hope, *supra* note 27.

[38] *See, e.g.*, Qianqian Xiea et al., FinBen: An Holistic Financial Benchmark for Large Language Models 1 (June 19, 2024) (unpublished manuscript), https://arxiv.org/pdf/2402.12659 [https://perma.cc/Z9DB-M5MM] (measuring various advances in financial tasks using AI and large language models).

B. *Market Manipulation and Financial Misconduct with AI*

Market manipulation and financial misconduct have been an existential fact and risk of finance.[39] Since there have been markets, there have been attempts to manipulate them.[40] The U.S. Supreme Court, in the landmark securities law case, *Santa Fe Industries v. Green*, opined that market manipulation "refers generally to practices, such as wash sales, matched orders, or rigged prices, that are intended to mislead investors by artificially affecting market activity."[41] Manipulated and distorted markets not only impact the prices and participants in one market but also have important implications for capital allocation, investments, and savings in other markets and the economy as a whole.[42] AI is but the latest and arguably most consequential means for bad actors to engage in market manipulation and financial misconduct to generate illicit gains during a period of increasing methods for financial misdeeds, a resurgence of retail investors, and the growth of autonomous financial programs.[43]

1. *Methods of Manipulation and Misconduct*

While the benefits of AI in finance are real, significant, and many, so are the risks.[44] Chief among these risks are the unprecedented methods for market

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[39] MARKHAM, *supra* note 12, at 9–14.

[40] *See* Tom C.W. Lin, *The New Market Manipulation*, 66 EMORY L.J. 1253, 1280 (2017) ("Market manipulation, broadly defined, has existed since the infancy of financial markets.").

[41] Santa Fe Indus., Inc. v. Green, 430 U.S. 462, 476 (1977).

[42] *See, e.g.*, Merritt B. Fox, Lawrence R. Glosten & Gabriel V. Rauterberg, *The New Stock Market: Sense and Nonsense*, 65 DUKE L.J. 191, 196 (2015) ("The performance of the equities market has important effects on the efficiency with which goods and services are produced in our economy and on the real economy's rate of growth. Equities also play a vital role as a place for ordinary individuals to invest their savings.").

[43] Ellen Sheng, *Generative AI Financial Scammers Are Getting Very Good at Duping Work Email*, CNBC (Feb. 14, 2024), https://www.cnbc.com/2024/02/14/gen-ai-financial-scams-are-getting-very-good-at-duping-work-email.html [https://perma.cc/RDU8-T2YE].

[44] For a wider discussion on the risks and rewards of AI and other earlier technologies in finance, see, for example, DANIELA RUS & GREGORY MONE, THE MIND'S MIRROR: RISK AND REWARD IN THE AGE OF AI 5–8, 163–72 (2024); Alessio Azzutti, *AI Trading and the Limits of EU Law Enforcement in Deterring Market Manipulation*, 45 COMPUT. L. & SEC. REV. 1, 3–6 (2022) (describing various types of market manipulation risk arising from the use of AI in trading of financial instruments); Kaminski, *supra* note 12, at 1354–57; Robert DeYoung, *Safety, Soundness, and the Evolution of the U.S. Banking Industry*, FED. RSRV. BANK OF ATLANTA ECON. REV., 1st & 2nd Quarters 2007, at 41, 42–44; and Loretta J. Mester, Commentary, *Some Thoughts on the Evolution of the Banking System and the Process of Financial Intermediation*, FED. RSRV. BANK OF ATLANTA ECON. REV., 1st & 2nd Quarters 2007, at 67, 67–72 (discussing how, while "[t]echnological and financial innovation . . . [have] increased competition" and "the geographic scope of markets," industry consolidation has also "present[ed] some conundrums").

manipulation and financial misconduct with AI.[45] Given the ease of use and growing proliferation of AI technologies, an axiom of the marketplace going forward will be the following: anything that can be manipulated with AI will be manipulated with AI.[46] Not surprisingly as a result of this growth, AI-driven applications and software have made it cheaper and easier for bad actors to engage in market manipulation and financial misconduct on a larger scale, especially with those methods involving information and misinformation.[47] There is even a dark web version of OpenAI's ChatGPT called FraudGPT—designed to create fake content for fraud and cyberattacks.[48]

Many forms of market manipulation involve information and misinformation.[49] A market participant with an illicit informational advantage exploits that edge to distort the price of an instrument to the detriment of others in the marketplace.[50] Consider the classic pump-and-dump schemes, depicted in popular films like *The Wolf of Wall Street*, whereby unscrupulous brokers use cold calls to spread misinformation to artificially inflate and sell worthless securities to unsuspecting, innocent investors.[51] Another form of informational market manipulation is benchmark manipulation, whereby market participants manipulate a key financial benchmark, thereby leading to distortions of instruments pegged to that benchmark.[52] A prominent example of benchmark

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[45] *See, e.g.*, Yonathan Arbel, Matthew Tokson & Albert Lin, *Systemic Regulation of Artificial Intelligence*, 56 ARIZ. ST. L.J. 545, 559–60 (2024) (describing various ways AI is "used to defraud individuals").

[46] *See, e.g.*, Dwork & Minow, *supra* note 11, at 315–16 (discussing problems of informational integrity resulting from the growth of AI).

[47] *See, e.g.*, Arbel, Tokson & Lin, *supra* note 45, at 560 ("One of the chief contributions of AI to the fraudulent enterprise is scale. AI will allow attackers to cast a much wider net by cutting the cost of interacting with each potential mark.").

[48] *See* Sheng, *supra* note 43; Zac Amos, *What Is FraudGPT?*, HACKERNOON (Aug. 11, 2023), https://hackernoon.com/what-is-fraudgpt [https://perma.cc/7ND8-97LV].

[49] *See* Azzutti, *supra* note 44, at 3–6 (describing different strategies and scenarios of market manipulation using AI); Lin, *supra* note 40, at 1281–94 (discussing various forms of market manipulation).

[50] *See* James Wm. Moore & Frank M. Wiseman, *Market Manipulation and the Exchange Act*, 2 U. CHI. L. REV. 46, 50 (1934) ("The term 'manipulation' may, in short, be applied to any practice which has as its purpose the deliberate raising, lowering or pegging of security prices. . . . Manipulation leads to an artificial and controlled price." (footnote omitted)); Chester Spatt, *Security Market Manipulation*, 6 ANN. REV. FIN. ECON. 405, 407 (2014) ("The investor in a classic manipulation is attempting to influence artificially the price as a way to gain potential advantage.").

[51] *See* Spatt, *supra* note 50, at 408 ("[A] pump and dump involves a trader 'pumping' up the price of a company by spreading false information to many unsophisticated (often retail) investors to push up the share prices and then 'dumping' the trader's shares."); *Pump and Dump Schemes*, U.S. SEC. & EXCH. COMM'N, http://www.sec.gov/answers/pumpdump.htm [https://perma.cc/F9ZA-LB54] ("Once the fraudsters dump their shares and stop hyping the stock, the stock price typically falls and investors lose money."); THE WOLF OF WALL STREET (Paramount Pictures 2013).

[52] Andrew Verstein, *Benchmark Manipulation*, 56 B.C. L. REV. 215, 217–18 (2015).

distortion is the London Interbank Offered Rate (LIBOR) scandal, which involved a group of bankers colluding to submit false bids to manipulate the benchmark rate to generate illicit gains for derivatives and other instruments pegged to LIBOR.[53] Benchmark distortion is highly problematic in today's marketplace where some of the most popular and widely held securities are exchange-traded and index funds tied directly to benchmarks like the Nasdaq Composite Index and the S&P 500 Index.[54] With AI programs, generating highly convincing fake news or data that can move individual stocks or impact key benchmarks is just a few taps away on one's phone.[55]

Furthermore, advances in AI technology have allowed autonomous financial systems to engage in market manipulation and financial misconduct through more sophisticated methods of pinging and spoofing, where machines are programmed to fool other machines to generate illicit gains.[56] Pinging is a form of market manipulation whereby significant volumes of small orders for a particular financial instrument or security are submitted and cancelled in fractions of a second by an AI-powered program to induce other machines in the marketplace to react to their "pings" and disclose their trading intentions to the pinging party.[57] The pinging party gains this valuable counterparty insight at little to no risk because its orders are not bona fide and often cancelled prior to execution.[58]

Similarly, spoofing is a form of market misconduct whereby AI-powered machines place fraudulent orders for a financial instrument or security at prices outside the current bona fide limits to trigger or bait the AI-powered systems of other market participants to distort price discovery in a manner favorable to the

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[53] *See* STEPHEN BLYTH, AN INTRODUCTION TO QUANTITATIVE FINANCE 27 (2014) (explaining LIBOR); DAVID ENRICH, THE SPIDER NETWORK: THE WILD STORY OF A MATH GENIUS, A GANG OF BACKSTABBING BANKERS, AND ONE OF THE GREATEST SCAMS IN FINANCIAL HISTORY 4–6 (2017) (chronicling the LIBOR scandal); Michael Corkery & Ben Protess, *Rigging of Foreign Exchange Market Makes Felons of Top Banks*, N.Y. TIMES (May 20, 2015), https://www.nytimes.com/2015/05/21/business/dealbook/5-big-banks-to-pay-billions-and-plead-guilty-in-currency-and-interest-rate-cases.html.

[54] *See* GARY L. GASTINEAU, THE EXCHANGE-TRADED FUNDS MANUAL 108–10 (2d ed. 2010) (discussing widely held exchange-traded funds benchmarked to key stock market indexes); Kent Thune, *Most Popular ETFs by AUM*, ETF.COM, https://www.etf.com/etf-education-center/etf-basics/most-popular-etfs [https://perma.cc/PX35-5A9X].

[55] *See* Maria Cristina Arcuri, Gino Gandolfi & Ivan Russo, *Does Fake News Impact Stock Returns? Evidence from US and EU Stock Markets*, J. ECON. & BUS., May–June 2023, Article No. 106130, at 1–2 (finding that fake news has short-term impact on equity markets).

[56] *See* Lin, *supra* note 40, at 1288–89; Takanobu Mizuta, *Can an AI Perform Market Manipulation at Its Own Discretion? A Genetic Algorithm Learns in an Artificial Market Simulation*, IEEE SYMP. SERIES ON COMPUTATIONAL INTEL. 407, 407, 411 (2020).

[57] *See* Gregory Scopino, *The (Questionable) Legality of High-Speed "Pinging" and "Front Running" in the Futures Markets*, 47 CONN. L. REV. 607, 611–14 (2015).

[58] BROWN, *supra* note 12, at 113.

spoofing party.[59] The spoofing party thus distorts the ordinary price discovery process by placing bad-faith orders for the purpose of manipulating honest participants in the marketplace without any bona fide intention to fulfill them.[60] Such schemes can be coupled with short-selling by the spoofing party to generate further downward momentum for the stock and more illicit gains from the market manipulation.[61]

Both pinging and spoofing are made possible and profitable by the rise of autonomous, high-speed, AI-driven financial technology.[62] This is because pinging, spoofing, and other similar methods of financial misconduct work best with high-speed submissions and cancellations of voluminous orders measured in velocities of fractions of a second.[63] Traditional human traders and brokers who execute orders and trades measured in minutes or hours simply could not carry out these illicit schemes in a profitable manner before being exposed by their counterparties.[64] High-speed AI-driven platforms, however, can execute these schemes to gain fractions of a penny per trade to the tune of billions of dollars in profits by taking advantage of counterparties with slower execution speeds and inferior codes.[65] As AI technologies have advanced in recent years, similar market manipulation strategies of the recent past will likely also advance and become more sophisticated.

Regardless of the method of market manipulation or financial misconduct, AI and related technologies have generally made it exponentially cheaper and

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[59] *See* Nw. Biotherapeutics, Inc. v. Canaccord Genuity LLC, 22 Civ. 10185, 2023 WL 9102400, at *2 (S.D.N.Y. Dec. 29, 2023) (describing spoofing as "typically utilizing high-frequency trading algorithms, creat[ing] the illusion of excess supply or demand by placing so-called 'Baiting Orders'"); Commodity Exchange Act, 7 U.S.C. § 6c(a)(5)(C) (defining spoofing); François-Serge Lhabitant & Greg N. Gregoriou, *High-Frequency Trading: Past, Present, and Future*, *in* HANDBOOK OF HIGH FREQUENCY TRADING 155, 161 (Greg N. Gregoriou ed., 2015) (explaining the underlying mechanics of spoofing).

[60] To be sure, depending on the intentions of the participants, some of these illicit trading practices can seem very similar to legitimate, good-faith trading strategies. *See, e.g.*, Massimiliano Marzo, *Designing a Trading Market*, *in* MARKET MICROSTRUCTURE IN EMERGING AND DEVELOPED MARKETS 159, 171 (H. Kent Baker & Halil Kiymaz eds., 2013).

[61] *See, e.g.*, Harrington Glob. Opportunity Fund, Ltd. v. CIBC World Mkts. Corp., 585 F. Supp. 3d 405, 412 (2022) (describing how spoofing schemes can be paired with short-selling to manipulate stock prices for illicit gains).

[62] *See* Scopino, *supra* note 57, at 610–16; Carol Clark & Rajeev Ranjan, *How Do Broker-Dealers/Futures Commission Merchants Control the Risks of High Speed Trading?*, FED. RSRV. BANK OF CHI., June 2012, at 1, 3.

[63] *See* Scopino, *supra* note 57, at 613–14; Clark & Ranjan, *supra* note 62, at 3.

[64] *See* James Surowiecki, *New Ways to Crash the Market*, NEW YORKER (May 11, 2015), http://www.newyorker.com/magazine/2015/05/18/new-ways-to-crash-the-market ("A human trader would never be able to quickly synthesize all the information in the order book, but a bot can.").

[65] *See, e.g.*, MICHAEL LEWIS, FLASH BOYS: A WALL STREET REVOLT 233 (2014).

easier to engage in informational market manipulation.[66] AI-powered software has made it easier for market participants to generate and widely disseminate market-sensitive information and misinformation.[67] In times past, significant technological resources were required to create and distribute market-moving information, but technological advances have made this possible today for anyone with a laptop or phone.[68] Free and fairly inexpensive AI programs can generate written content of various styles and formats, including those akin to professional financial research.[69] Once created, AI-generated content can be distributed widely and amplified on social media via AI-driven bots and software programs.[70] With AI, misinformation can readily become the information of the day, and manipulative information can readily become market-moving information.[71] Market manipulation with AI is particularly troubling in our modern marketplace because of the resurgence of retail investors and the rise of autonomous financial programs.

2. *A Resurgence of Retail Investors*

Market manipulation with AI can significantly harm the millions of retail investors with hundreds of billions of dollars' worth of assets in the marketplace.[72] Retail investors have played an increasingly influential role in

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[66] *See, e.g.*, David Klepper & Ali Swenson, *AI Presents Political Peril for 2024 with Threat to Mislead Voters*, AP (May 14, 2023), https://apnews.com/article/artificial-intelligence-misinformation-deepfakes-2024-election-trump-59fb51002661ac5290089060b3ae39a0 [https://perma.cc/8P5A-LBJS]; Stuart A. Thompson, *Making Deepfakes Gets Cheaper and Easier Thanks to A.I.*, N.Y. TIMES (Mar. 12, 2023), https://www.nytimes.com/2023/03/12/technology/deepfakes-cheapfakes-videos-ai.html; Alexander Kuper, *Generative AI and Disinformation*, BIPARTISAN POL'Y CTR. (Aug. 23, 2023), https://bipartisanpolicy.org/blog/generative-ai-and-disinformation [https://perma.cc/CH6V-2W9P].

[67] *See, e.g.*, NOAH GIANSIRACUSA, HOW ALGORITHMS CREATE AND PREVENT FAKE NEWS 17 (2021).

[68] *See* Tiffany Hsu & Stuart A. Thompson, *Disinformation Researchers Raise Alarms About A.I. Chatbots*, N.Y. TIMES, https://www.nytimes.com/2023/02/08/technology/ai-chatbots-disinformation.html (June 20, 2023).

[69] *See* Will Knight, *It Costs Just $400 to Build an AI Disinformation Machine*, WIRED (Aug. 29, 2023), https://www.wired.com/story/400-dollars-to-build-an-ai-disinformation-machine [https://perma.cc/SS92-48TR].

[70] *See* GIANSIRACUSA, *supra* note 67, at 175–216.

[71] *See, e.g.*, Amy Chozick & Nicole Perlroth, *Twitter Speaks, Markets Listen and Fears Rise*, N.Y. TIMES (Apr. 28, 2013), https://www.nytimes.com/2013/04/29/business/media/social-medias-effects-on-markets-concern-regulators.html (describing how information on social media can impact the stock market).

[72] *See* Alexandra Semenova, *Retail Investors Are Pouring a Record $1.5 Billion Per Day into the Stock Market*, YAHOO! FIN. (Feb. 16, 2023), https://finance.yahoo.com/news/retail-investors-record-inflows-us-stock-market-193801422.html [https://perma.cc/7EVY-XDJT]; Karl Ehrsam, Irena Gecas-McCarthy, Robert Walley & Jill Gregorie, *The Rise of Newly Empowered Individual Investors*, WALL ST. J. (June 16, 2021), https://deloitte.wsj.com/

our financial markets in recent years.[73] This was made particularly evident during the meme stock frenzy in 2020 and 2021, when retail investors caused a number of stocks to hit record prices based on information circulated on social media.[74] Free and easy-to-use trading apps, coupled with access to numerous channels for information, render contemporary retail investors particularly vulnerable to market manipulation with AI.[75]

Manipulative information and misinformation generated and proliferated by AI on social media platforms can induce already active retail investors to make terrible trades that they otherwise would not make in the absence of such misinformation.[76] This is especially problematic in today's highly concentrated marketplace, where the stocks of just seven companies in 2024—Alphabet, Amazon, Apple, Meta, Microsoft, Nvidia, or Tesla—accounted for most of the gains of the entire market.[77] If the stock of any one of those companies was subjected to misinformation, it could have serious deleterious effects on the entire market, resulting in significant losses for many retail and institutional investors.

3. *The Rise of Autonomous Financial Programs*

Market manipulation and financial misconduct with AI are particularly worrisome in our modern marketplace due to the rise of autonomous financial programs.[78] So much of modern finance is driven by smart machines with little to no human intervention.[79] Even prior to the latest iteration of AI, smart machines have for years traded hundreds of billions of dollars in equity and debt

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articles/the-rise-of-newly-empowered-individual-investors-01623870131 [https://perma.cc/LL8S-TVN9]; Krishan Arora, *The Rise of the Retail Investor*, FORBES (Nov. 4, 2022), https://www.forbes.com/sites/forbesagencycouncil/2022/11/04/the-rise-of-the-retail-investor [https://perma.cc/Z4PV-PH7V].

[73] *See* Sue S. Guan, *Meme Investors and Retail Risk*, 63 B.C. L. REV. 2051, 2054–56 (2022); James Fallows Tierney, *Investment Games*, 72 DUKE L.J. 353, 373–77 (2022).

[74] Fisch, *supra* note 12, at 1802–03.

[75] *See* Sergio Alberto Gramitto Ricci & Christina M. Sautter, *Corporate Governance Gaming: The Collective Power of Retail Investors*, 22 NEV. L.J. 51, 67–73 (2021) (discussing new class of retail investors that use technology and information to move markets); U.S. SEC. & EXCH. COMM'N, STAFF REPORT ON EQUITY AND OPTIONS MARKET STRUCTURE CONDITIONS IN EARLY 2021, at 6–9 (2021), https://www.sec.gov/files/staff-report-equity-options-market-struction-conditions-early-2021.pdf [https://perma.cc/746C-HATB].

[76] *See* Tierney, *supra* note 73, at 356–67, 377.

[77] *See* Matt Phillips, *The S&P's Top Stocks Haven't Been This Concentrated in 50 Years*, AXIOS (Feb. 5, 2024), https://www.axios.com/2024/02/05/sp500-magnificent-seven-stock-market [https://perma.cc/HU96-MFZX]; Wayne Duggan, *Magnificent 7 Stocks: What Are They and How They Dominate the Market*, U.S. NEWS, https://money.usnews.com/investing/articles/magnificent-7-stocks-explainer [https://perma.cc/7ACC-C9UJ] (Jan. 7, 2025).

[78] *See* GREGORY SCOPINO, ALGO BOTS AND THE LAW: TECHNOLOGY, AUTOMATION, AND THE REGULATION OF FUTURES AND OTHER DERIVATIVES 7–8 (2020).

[79] *See* Fletcher & Le, *supra* note 25, at 301–02.

with other smart machines at speeds measured in milliseconds.[80] Automated computer programs assess and manage trillions of dollars' worth of assets for the largest wealth managers in the world.[81] Autonomous bots track, detect, and flag millions of transactions for irregularities, fraud, and compliance in this high-speed environment.[82] All of this autonomous financial technology depends on powerful computer algorithms powered by accurate data and good information.[83] Once the integrity of the data and information is corrupted by misinformation and misconduct, these smart machines may no longer seem so intelligent, profitable, or socially beneficial for many market participants and observers.[84]

Additionally, AI has made it much easier for smart machines to fool other smart machines before any human can intervene to stop the significant financial losses and social harm.[85] AI has also made it possible for smart machines to exploit vulnerabilities in the market and weaknesses in counterparties for profit without regard for any legal norms or concerns.[86] AI has made it possible for smart machines to learn to cheat, collude, and manipulate financial markets in ways that humans could not previously devise or execute because they lacked the data processing powers of contemporary self-learning machines.[87]

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[80] *See* Charles Duhigg, *Stock Traders Find Speed Pays, in Milliseconds*, N.Y. TIMES (July 23, 2009), http://www.nytimes.com/2009/07/24/business/24trading.html ("[Algorithmic computer programs] can spot trends before other investors can blink, changing orders and strategies within milliseconds.").

[81] *See* Damir Tokic, *BlackRock Robo-Advisor 4.0: When Artificial Intelligence Replaces Human Discretion*, 27 STRATEGIC CHANGE 285, 286–89 (2018).

[82] *See* Concept Release on Risk Controls and System Safeguards for Automated Trading Environments, 78 Fed. Reg. 56542, 56573 app. 2 (Sept. 12, 2013) (to be codified at 17 C.F.R. ch. 1) ("Automated trading systems, including high frequency traders, enter the market and execute trades in a matter of milliseconds without human involvement.").

[83] *See* Sonia K. Katyal, *Private Accountability in the Age of Artificial Intelligence*, 66 UCLA L. REV. 54, 58–59 (2019) (discussing the importance of data behind algorithmic decision-making in connection with AI); *see also* PEDRO DOMINGOS, THE MASTER ALGORITHM 1–3 (2015) (enumerating various benefits of machine learning based on large troves of data).

[84] *See generally* MICHAEL CHUI ET AL., MCKINSEY & CO., NOTES FROM THE AI FRONTIER: APPLYING AI FOR SOCIAL GOOD (2018).

[85] *See* CATHY O'NEIL, WEAPONS OF MATH DESTRUCTION 3 (2016) ("[M]any of these models encoded human prejudice, misunderstanding, and bias into the software systems that increasingly managed our lives.").

[86] *See generally* Mizuta, *supra* note 56 (studying how AI can autonomously learn to manipulate financial markets through algorithmic trading).

[87] *See* Alessio Azzutti, Wolf-Georg Ringe & H. Siegfried Stiehl, *Machine Learning, Market Manipulation, and Collusion on Capital Markets: Why the "Black Box" Matters*, 43 U. PA. J. INT'L L. 79, 82–83 (2021).

In sum, market misinformation, manipulation, and misconduct with AI present some of the chief dangers of the modern marketplace. To be sure, market manipulation and financial misconduct have always existed and have evolved with changes in society and technology. The latest emerging iteration of AI presents a particularly noteworthy threat because it represents a social and technological shift, a shift so potentially significant that it accordingly changes the magnitude of market misinformation, manipulation, misconduct, and impact.

C. *Financial Deepfakes*

While the means and methods for manipulation and misconduct with AI are limited only by the devious imaginations of bad actors, the use of AI-generated deepfakes warrants additional scrutiny due to its pernicious impact on financial markets and investors.[88] "AI deepfakes" involve using AI to create highly convincing inauthentic images, videos, audio, and written content that are incredibly difficult to discern from reality and truth, thereby eroding trust generally.[89] In recent years, society has witnessed the use of AI deepfakes to generate heartwarming videos, creative music, and funny images.[90] At the same time, society has witnessed the use of AI deepfakes to confuse voters, degrade women, embezzle money, and cause ethnic conflict.[91] Between 2022 and 2023 alone, it was estimated that incidents of deepfake-related misconduct increased by over 1,000%.[92] Given the growth and wide-ranging uses of AI deepfakes as misinformation, it should come as no surprise that those who seek illicit profit

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88 *See generally* MATTHEW MILLER, KPMG, DEEPFAKES: REAL THREAT (2023), https://kpmg.com/kpmg-us/content/dam/kpmg/pdf/2023/deepfakes-real-threat.pdf [https://perma.cc/4364-Z5FE] (reporting on the various threats deepfakes pose to businesses).

89 *See* Mary Anne Franks & Ari Ezra Waldman, *Sex, Lies, and Videotape: Deep Fakes and Free Speech Delusions*, 78 MD. L. REV. 892, 893 (2019) ("[D]eep fakes erode the capacity of the public to discern truth from falsity."); NAT'L SEC. COMM'N ON A.I., *supra* note 15, at 604 (defining deepfakes as "[c]omputer-generated video or audio (particularly of humans) so sophisticated that it is difficult to distinguish from reality.").

90 *See* Karen Hao & Will Douglas Heaven, *The Year Deepfakes Went Mainstream*, MIT TECH. REV. (Dec. 24, 2020), https://www.technologyreview.com/2020/12/24/1015380/best-ai-deepfakes-of-2020.

91 *See* Chesney & Citron, *supra* note 12, at 1772–85 (enumerating and discussing various harms that can be perpetuated by deepfakes); Emily Flitter & Stacy Cowley, *Voice Deepfakes Are Coming for Your Bank Balances*, N.Y. TIMES (Aug. 30, 2023), https://www.nytimes.com/2023/08/30/business/voice-deepfakes-bank-scams.html ("[T]he speed of technological development, the falling costs of generative artificial intelligence programs and the wide availability of recordings of people's voices on the internet have created the perfect conditions for voice-related A.I. scams.").

92 *Sumsub Research: Global Deepfake Incidents Surge Tenfold from 2022 to 2023*, SUMSUB (Nov. 28, 2023), https://sumsub.com/newsroom/sumsub-research-global-deepfake-incidents-surge-tenfold-from-2022-to-2023 [https://perma.cc/2P84-2JW8].

through market manipulation also use it to create financial deepfakes for their benefit.[93]

With the use of relatively inexpensive and accessible AI tools, bad actors can easily produce highly convincing fake images, videos, audio, and written content to move the market.[94] Video and audio deepfakes of business leaders can be generated to move a particular stock before they can be authenticated and rebuked.[95] Imagine the impact on the marketplace if there were a fake audio clip of Tim Cook, the longtime chief executive of Apple, speaking about the imminent launch of a highly rumored Apple car, driving the stock up only for it to be revealed that the clip was not real. Similarly, bad actors can use AI to generate highly convincing emails, videos, social media posts, regulatory filings, draft agreements, resignation letters, and other market-moving content for their own malicious ends.[96] No business or executive is safe from deepfakes.[97] In fact, in 2024, Taylor Swift—a global superstar and one of the most influential individuals in the world—was subjected to an onslaught of disgusting pornographic deepfakes aimed at damaging her reputation.[98]

The growth in financial deepfakes could also erode confidence in the integrity of the marketplace as investors become wary of trusting financial information for fear of being fooled, manipulated, or losing money if the information turns out to be true.[99] While there are many means of rapidly

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[93] *See generally* JON BATEMAN, DEEPFAKES AND SYNTHETIC MEDIA IN THE FINANCIAL SYSTEM: ASSESSING THREAT SCENARIOS (2020), https://carnegie-production-assets.s3.amazonaws.com/static/files/Bateman_FinCyber_Deepfakes_final.pdf [https://perma.cc/UKK8-LKKJ].

[94] *See* NAT'L SEC. COMM'N ON A.I., *supra* note 15, at 22 ("'Deepfake' capabilities can be easily downloaded and used by anyone."); Rhiannon Williams, *Humans May Be More Likely to Believe Disinformation Generated by AI*, MIT TECH. REV. (June 28, 2023), https://www.technologyreview.com/2023/06/28/1075683/humans-may-be-more-likely-to-believe-disinformation-generated-by-ai.

[95] *See* Chesney & Citron, *supra* note 12, at 1775 ("Deep fakes can be used to sabotage business competitors.").

[96] *See id.* ("Deep-fake videos could show a rival company's chief executive engaged in any manner of disreputable behavior, from purchasing illegal drugs to hiring underage prostitutes to uttering racial epithets to bribing government officials. Deep fakes could be released just in time to interfere with merger discussions or bids for government contracts."); Sheng, *supra* note 43.

[97] *See* Sheng, *supra* note 43.

[98] *See, e.g.*, Margi Murphy, *Taylor Swift Deepfakes Originated from AI Challenge, Report Says*, BLOOMBERG L. (Feb. 5, 2024), https://news.bloomberglaw.com/artificial-intelligence/taylor-swift-deepfakes-originated-from-ai-challenge-report-says; Kate Conger & John Yoon, *Explicit Deepfake Images of Taylor Swift Elude Safeguards and Swamp Social Media*, N.Y. TIMES (Jan. 26, 2024), https://www.nytimes.com/2024/01/26/arts/music/taylor-swift-ai-fake-images.html.

[99] *See* Chesney & Citron, *supra* note 12, at 1779 ("Deep fakes will erode trust in a wide range of both public and private institutions and such trust will become harder to maintain."); Andrew Ross Sorkin et al., *Investors Fear Bank Contagion, Despite a Sweeping Rescue*

generating and amplifying financial deepfakes, there remain very limited means to expose and rebuke them.[100] Because of the high velocity of the modern marketplace, a financial deepfake can cause serious losses and harms before it is disproven.

When in doubt about a company's future and in the face of an onslaught of highly convincing misinformation, it is not unreasonable for investors to simply sell and exit their positions. In 2023, the online speculations of a hypothetical bank run helped fuel an actual offline bank run on Silicon Valley Bank, the then-already-troubled sixteenth-largest bank in the United States and a key institution in the technology industry.[101] One of the leading contributing factors to the run was the abundance of rumors and misinformation online about the fragile bank's demise, which were amplified by its substantial clientele in the technology industry, many of whom ultimately turned these rumors into a self-fulfilling prophecy.[102]

Because AI technologies will only continue to advance and get more sophisticated, the quality and corresponding challenges posed by financial deepfakes will also advance. While federal legislation addressing deepfakes has been elusive, many states, led by California, have taken legislative action to address some of these emerging challenges.[103]

Plan, N.Y. TIMES (Mar. 13, 2023), https://www.nytimes.com/2023/03/13/business/dealbook/first-republic-contagion-svb-banks.html ("Bank runs are even more dangerous in the age of social media. Confidence can evaporate faster than ever when misinformation can spread in a matter of minutes, and a single tweet can send customers fleeing.").

100 *See* Chesney & Citron, *supra* note 12, at 1787–88 (describing various challenges to detecting deepfakes).

101 Vivian Giang, *Banking Turmoil: What We Know*, N.Y. TIMES (Mar. 15, 2023), https://www.nytimes.com/article/svb-silicon-valley-bank-explainer.html; Ken Sweet, *One of Silicon Valley's Top Banks Fails; Assets Are Seized*, AP (Mar. 11, 2023), https://apnews.com/article/svb-fed-bonds-rates-banks-inflation-a24b28b3caeede91c76cd120aa9b7966 [https://perma.cc/KK89-PRLJ].

102 *See* Kevin Roose, *3 Lessons from Silicon Valley Bank's Failure*, N.Y. TIMES (Mar. 11, 2023), https://www.nytimes.com/2023/03/11/technology/silicon-valley-bank-failure-lessons.html ("Once a few people in tech raised questions about the firm's solvency, Slack channels and Twitter feeds lit up with dire warnings from venture capitalists, and soon many people were panicking.").

103 *See, e.g.*, CAL. GOV'T CODE § 11547.5 (West 2023); *Governor Newsom Signs Bills to Combat Deepfake Election Content*, GOVERNOR OF CAL. (Sept. 17, 2024), https://www.gov.ca.gov/2024/09/17/governor-newsom-signs-bills-to-combat-deepfake-election-content [https://perma.cc/2SUX-8EBH]; Trân Nguyễn, *California Laws Cracking Down on Election Deepfakes by AI Face Legal Challenges*, AP (Sept. 18, 2024), https://apnews.com/article/california-artificial-intelligence-deepfakes-election-3cf47301380b01ab35925a1c0a78171f [https://perma.cc/4K7V-QH64]; Stuart A. Thompson, *California Passes Election 'Deepfake' Laws, Forcing Social Media Companies to Take Action*, N.Y. TIMES (Sept. 17, 2024), https://www.nytimes.com/2024/09/17/technology/california-deepfakes-law-social-media-newsom.html.

D. *The Liar's Dividend and the Truthteller's Tax*

The proliferation of AI-powered misinformation and persistent market misconduct can further erode trust so significantly that investors and the investing public become more skeptical of all information, including authentic information.[104] As a result of this erosion in trust, firms and regulators end up expending more resources to rebuild trust in the marketplace and convince the investing public of the veracity of their truthful information and disclosures.[105] Because investments and economic growth thrive in environments of high trust, the erosion of trust caused by AI-powered market manipulation and financial misconduct would ultimately have a negative effect on our individual and collective prosperity.[106]

Truth and facts are already under attack in modern life by ideology, bias, and power.[107] As our public spaces and private markets become further polluted with AI-driven deepfakes and misinformation, our environments for trade and transactions will lose greater trust and suffer from "truth decay."[108] Trust in business is highly dependent on the social and economic environment of the informational exchange or commercial transaction.[109] Meaningful business transactions simply do not happen without some level of trust. Economic Nobel Laureate Kenneth Arrow wrote that "[v]irtually every commercial transaction has within itself an element of trust, certainly any transaction conducted over a period of time."[110]

Over time, as the public becomes more acculturated and educated about deepfakes and other forms of misinformation—financial and otherwise—people could also become more inclined to believe denials by bad actors when presented with truthful information about their misdeeds.[111] For instance, consider the following scenario: a rogue nation-state spreads misinformation

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104 *See* Chesney & Citron, *supra* note 12, at 1785 ("[A] skeptical public will be primed to doubt the authenticity of real audio and video evidence. This skepticism can be invoked just as well against authentic as against adulterated content.").

105 *See* Paul J. Zak & Stephen Knack, *Trust and Growth*, 111 ECON. J. 295, 296 (2001) ("Because trust reduces the cost of transactions ([i.e.,] less time is spent investigating one's broker), high trust societies produce more output than low trust societies.").

106 *See id.* at 317 ("[I]nvestment and growth improve with trust."); Ira Kalish, Michael Wolf & Jonathan Holdowsky, *The Link Between Trust and Economic Prosperity*, DELOITTE: INSIGHTS (May 20, 2021), https://www.deloitte.com/us/en/insights/economy/connecting-trust-and-economic-growth.html [https://perma.cc/R7YS-JEYE] ("[T]he positive relationship between trust and the macroeconomy is well-known.").

107 ALEX EDMANS, MAY CONTAIN LIES: HOW STORIES, STATISTICS, AND STUDIES EXPLOIT OUR BIAS—AND WHAT WE CAN DO ABOUT IT 9–10 (2024).

108 *See* JENNIFER KAVANAGH & MICHAEL D. RICH, TRUTH DECAY x–xv (2018) (explaining truth decay and its causes).

109 *See* Zak & Knack, *supra* note 105, at 296 ("[T]rust depends on the social, economic and institutional environments in which transactions occur.").

110 Kenneth J. Arrow, *Gifts and Exchanges*, 1 PHIL. & PUB. AFFS. 343, 357 (1972).

111 Chesney & Citron, *supra* note 12, at 1785.

about a major American bank that leads to short-term volatility in the American marketplace. Subsequently, when credible, authentic video evidence is publicized about the origins of the financial misconduct, the leader of the rogue nation-state denies it and calls it fake news, and many in the public believe him because of the pervasiveness of deepfakes. Prominent legal scholars, Professors Bobby Chesney and Danielle Citron, termed this pernicious phenomenon "the liar's dividend."[112]

The liar's dividend has serious implications for the well-being of business and society because it depreciates the collective trust in the economy, and this trust is critical to investment, growth, and high standards of living.[113] A 2021 report from Deloitte found that "[c]ountries where businesses, governments, and other institutions have engendered more trust experience stronger per capita real GDP growth, a standard measure of economic prosperity."[114] As investors and the public grow more skeptical and less trusting of information, firms, and the market generally because of highly convincing AI-generated content, they are more likely to hoard capital, invest less, and withdraw from the marketplace.[115]

Perhaps more troubling, as misinformation proliferates, society will likely become more habituated to the falsehoods and engage in greater dishonesty as the distinctions between truth and lies become less meaningful and distinguishable to more people.[116] As the philosopher Hannah Arendt presciently noted in a 1967 *New Yorker* essay, "In other words, the result of a consistent and total substitution of lies for factual truth is not that the lies will now be accepted as truth, and the truth be defamed as lies, but that the sense by which we take our bearings in the real world—and the category of truth vs. falsehood is among the mental means to this end—is being destroyed."[117]

In an effort to discount and counteract the deleterious effects of the liar's dividend, regulators and firms would have to spend more time and resources to convince a suspicious and skeptical investing public of the veracity of their

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112 *Id.*

113 *See* Kalish, Wolf & Holdowsky, *supra* note 106 ("A lack of trust can make some investments appear too risky, which can lead to suboptimal investment allocations. For example, countries with low levels of trust tend to invest in projects with shorter time horizons. Investments with longer time horizons require more trust in workers to complete the project, in suppliers to get the necessary equipment, and in customers to continue to be there through the useful life of the investment." (footnote omitted)).

114 *Id.*

115 *See* Luigi Guiso, Paola Sapienza & Luigi Zingales, *Trusting the Stock Market*, 63 J. FIN. 2557, 2557–60 (2008) (finding that trust is a key contributing factor to investment decisions in the stock market); Kalish, Wolf & Holdowsky, *supra* note 106 ("Stronger trust in the financial sector makes capital more readily available . . . . Banks lacking trust face quicker savings withdrawals during times of distress. Evidence suggests that the first people to withdraw their deposits following the collapse of Lehman Brothers were those who had the least trust in their bank." (footnote omitted)).

116 *Cf.* TALI SHAROT & CASS R. SUNSTEIN, LOOK AGAIN: THE POWER OF NOTICING WHAT WAS ALWAYS THERE 102–08 (2024).

117 Hannah Arendt, *Truth and Politics*, NEW YORKER, Feb. 25, 1967, at 49, 78.

information and disclosures; and counterparties would have to expend additional transactional costs to verify information that otherwise would not be necessary in high-trust environments.[118] I call these additional expenditures and transactional costs "the truthteller's tax." This tax could be enormous if the public grows ever more skeptical and suspicious of financial (mis)information because psychological biases make changing someone's beliefs extremely difficult.[119] The truthteller's tax thus results in higher costs as firms and counterparties spend more money to rebuild trust and comply with regulatory attempts for the sake of ensuring adequate capital investments and economic productivity.[120] Many of these additional transactional costs are likely passed on to investors, consumers, and society writ-large.[121] Therefore, the proliferation of AI-powered market manipulation, misinformation, and misconduct—if left unchecked—could fundamentally undermine our economic system and hurt our personal and social well-being.[122]

III. SYSTEMIC RISKS, THREATS, AND ACCIDENTS

The rise of AI has led to many predictions about its significant benefits, but also its serious risks.[123] The invention of AI technologies and innovations is also the invention of AI risks and dangers.[124] As the philosopher Paul Virilio wrote, "To invent the sailing ship or the steamer is to invent the shipwreck. To invent

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118 *See* Kalish, Wolf & Holdowsky, *supra* note 106 ("Simply put, lacking trust can be expensive. Writing and enforcing contracts, monitoring worker and subcontractor behavior, and implementing security protocols cost money.").

119 *See* Chesney & Citron, *supra* note 12, at 1786 ("Cognitive bias will reinforce these unhealthy dynamics.").

120 *See* Alexander Lascaux, *Trust and Transaction Costs*, *in* COMPLEXITY AND THE ECONOMY 151, 156–57 (John Finch & Magali Orillard eds., 2005) (discussing how low trust leads counterparties in exchanges to incur more transaction costs); Kalish, Wolf & Holdowsky, *supra* note 106 ("Greater levels of trust can raise the quantity and quality of investments in physical and human capital. It can also boost productivity growth through more effective organizational adjustments.").

121 *See* Lascaux, *supra* note 120, at 151–54 (explaining the increasing transaction costs arising from lack of trust).

122 *See* Zak & Knack, *supra* note 105, at 317 ("[M]ore inclusive measures of well-being will be associated with trust in the same way that . . . investment and growth improve with trust."); Diane Coyle, *The Cost of Mistrust*, OECD YEARBOOK, 2013, at 73, 74 ("[T]he evaporation of trust in some key institutions is directly damaging for economic growth. The general distrust of business, not only banks but also spreading to energy companies, e-retailers, food producers and others, will tend to encourage unnecessary (as well as necessary) regulation and contribute to the climate of uncertainty holding back investment.").

123 *See, e.g.*, Bernhard Babel, Kevin Buehler, Adam Pivonka, Bryan Richardson & Derek Waldron, *Derisking Machine Learning and Artificial Intelligence*, MCKINSEY & CO. (Feb. 19, 2019), https://www.mckinsey.com/capabilities/risk-and-resilience/our-insights/derisking-machine-learning-and-artificial-intelligence.

124 Metz & Schmidt, *supra* note 10.

the train is to invent the rail accident of derailment. To invent the family automobile is to produce the pile-up on the highway."[125] Some of these predictions about AI foretell nightmare scenarios, including an apocalyptic end to humanity.[126] In 2023, a growing list of over 350 leading executives, scientists, and engineers signed on to a one-sentence statement from the Center for AI Safety that read, "Mitigating the risk of extinction from AI should be a global priority alongside other societal-scale risks such as pandemics and nuclear war."[127] Whether this ominous future will materialize remains uncertain, but real systemic risks and threats to the financial system already exist because of the proliferation of AI in the marketplace.[128] Market misinformation, manipulation, and misconduct are in many ways emblematic symptoms of these larger structural hazards and vulnerabilities posed by AI.[129]

A. *Systemic Market Risks*

The infusion of AI into our financial markets creates new, interrelated systemic risks involving speed and opacity. In the 2008 financial crisis, the key systemic risk was "too big to fail," a vulnerability related to large financial institutions and their potential deleterious impacts on the entire system should they become unsound.[130] With the wide adoption of AI technologies in finance, regulators, investors, and businesses must grow more watchful of the following emerging systemic risks: "too fast to stop" and "too opaque to understand."

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[125] PAUL VIRILIO, THE ORIGINAL ACCIDENT 10 (Julie Rose trans., Polity Press 2007) (2005) (emphasis omitted).

[126] *See* BOSTROM, *supra* note 12, at 115–18 (discussing the "existential catastrophe" that could arise from advances in AI); Kevin Roose, *A.I. Poses 'Risk of Extinction,' Industry Leaders Warn*, N.Y. TIMES (May 30, 2023), https://www.nytimes.com/2023/05/30/technology/ai-threat-warning.html; Metz, *supra* note 10.

[127] *Statement on AI Risk*, CTR. FOR AI SAFETY, https://www.safe.ai/statement-on-ai-risk [https://perma.cc/Q8M6-TEGM].

[128] *See* Tom C.W. Lin, *Artificial Intelligence, Finance, and the Law*, 88 FORDHAM L. REV. 531, 534 (2019) (highlighting various financial risks associated with AI).

[129] *See* Noam Kolt, *Algorithmic Black Swans*, 101 WASH. U. L. REV. 1177, 1182–83 (2024) (describing the "high-impact risks from AI" to society and markets).

[130] *See* PERMANENT SUBCOMM. ON INVESTIGATIONS, S. COMM. ON HOMELAND SEC. & GOVERNMENTAL AFFS., 112TH CONG., WALL STREET AND THE FINANCIAL CRISIS: ANATOMY OF A FINANCIAL COLLAPSE 15–22 (Comm. Print 2011) (discussing the systemic risk of "too-big-to-fail" institutions); ANDREW ROSS SORKIN, TOO BIG TO FAIL 538–39 (2009) (reporting on "too big to fail" institutions in connection with the financial crisis of 2008).

1. *Too Fast to Stop*

The wider adoption of AI and related technologies in the marketplace introduces a new systemic risk related to financial speed.[131] AI-powered acceleration of financial velocities creates a systemic risk of "too fast to stop," whereby misinformation, manipulation, and misconduct could destabilize the financial system before corrective or preventive measures can be implemented to stem the fallout.[132] Modern finance already moves at incredible speeds measured in fractions of a second.[133] AI-powered autonomous systems execute trades in milliseconds, literally thousands of times faster than the blink of an eye.[134] In the time it takes to read this sentence, billions of dollars could have been traded by high-speed, automated smart computers. It is estimated that AI-powered algorithmic trading accounts for over 60% of all equities transactions in the United States.[135] As AI and related technology advance, the speed of finance will undoubtedly accelerate.

This AI-induced financial acceleration often leads to more timely price discovery, greater liquidity, and better efficiencies for the marketplace when working properly during periods of stability.[136] However, during periods of confusion, distress, and panic, this high-tech accelerated velocity can lead to destabilizing volatility and calamity by rapidly increasing or decreasing liquidity as machines trade with one another.[137] It is not surprising that some of the most significant flash crashes and market volatility occurred during periods

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[131] *See* Lin, *supra* note 40, at 1296 ("[T]he unprecedented speed of many of today's transactions and trades makes it especially tough for regulators to pinpoint ongoing market manipulation schemes.").

[132] *See* Kirilenko & Lo, *supra* note 12, at 60 ("[A]utomated trading systems provide enormous economies of scale and scope in managing large portfolios, but trading errors can now accumulate losses at the speed of light before they're discovered and corrected by human oversight."); Lin, *supra* note 23, at 711–14.

[133] Lin, *supra* note 40, at 1296–97.

[134] *See* Frank J. Fabozzi, Sergio M. Focardi & Caroline Jonas, *High-Frequency Trading: Methodologies and Market Impact*, 19 REV. FUTURES MKTS. 7, 8–10 (2011).

[135] Oddmund Groette, *What Percentage of Trading Is Algorithmic? (Algo Trading Market Statistics)*, QUANTIFIED STRATEGIES (Apr. 7, 2024), https://www.quantifiedstrategies.com/what-percentage-of-trading-is-algorithmic [https://perma.cc/388C-EQDU].

[136] *See* Donald C. Langevoort & Robert B. Thompson, *"Publicness" in Contemporary Securities Regulation After the JOBS Act*, 101 GEO. L.J. 337, 347 (2013) ("Today, liquidity is now much more possible outside of traditional exchanges. In the new millennium, cheap information and low communication costs have expanded markets . . . ."); Jonathan Brogaard, Terrence Hendershott & Ryan Riordan, *High-Frequency Trading and Price Discovery*, 27 REV. FIN. STUD. 2267, 2267 (2014).

[137] *See* FRANK PARTNOY, WAIT: THE ART AND SCIENCE OF DELAY 43 (2012).

of widespread technology adaptation within the financial industry over the last few decades.[138]

In sum, speed, like size in financial crises past, will be a key systemic risk factor in a future crisis or significant market disruption with the rise of AI.[139] As a quotation often misattributed to Mark Twain demonstrates, "a lie can travel halfway around the world while the truth is still putting on its shoes"—and in the age of AI, financial misinformation can travel around the world before regulators can figure out what is true.[140]

2. *Too Opaque to Understand*

The wider adoption of AI in the marketplace introduces a new systemic risk related to opacity, which this Article terms "too opaque to understand."[141] The opacity of AI algorithms, often referred to as "black boxes," introduces a layer of complexity to understanding and regulating market dynamics.[142] Autonomous generative AI programs executing trades, allocating capital, and managing risks entirely on their own, without human actors being able to clearly understand or explain their models, are becoming more the norm in finance than the exception.[143] Once a machine is programmed to achieve some objective, the means by which it self-learns and self-operates on large data and large language models to achieve that objective—in legal or illegal ways—can often be a

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[138] *See, e.g.*, Louise Story & Graham Bowley, *Market Swings Are Becoming New Standard*, N.Y. TIMES (Sept. 11, 2011), https://www.nytimes.com/2011/09/12/business/economy/stock-markets-sharp-swings-grow-more-frequent.html; STAFFS OF THE U.S. COMMODITIES FUTURES TRADING COMM'N & U.S. SEC. & EXCH. COMM'N, FINDINGS REGARDING THE MARKET EVENTS OF MAY 6, 2010, at 1 (2010), https://www.sec.gov/news/studies/2010/marketevents-report.pdf [https:/perma.cc/E8VP-AAT8]. *See generally* Timothy Lavin, *Monsters in the Market*, ATLANTIC, July/Aug. 2010, https://www.theatlantic.com/magazine/archive/2010/07/monsters-in-the-market/308122; U.S. DEP'T OF TREASURY, BD. OF GOVERNORS OF FED. RSRV. SYS., FED. RSRV. BANK OF N.Y., U.S. SEC. & EXCH. COMM'N & U.S. COMMODITIES FUTURES TRADING COMM'N, JOINT STAFF REPORT: THE U.S. TREASURY MARKET ON OCTOBER 15, 2014 (2015), https://home.treasury.gov/system/files/276/joint-staff-report-the-us-treasury-market-on-10-15-2014.pdf [https://perma.cc/2CBT-5J3S].

[139] *See* Azzutti, Ringe & Stiehl, *supra* note 87, at 84 (discussing how AI-powered financial algorithms can "cause disruptions to the safe and orderly functioning of markets").

[140] Niraj Chokshi, *That Wasn't Mark Twain: How a Misquotation Is Born*, N.Y. TIMES (Apr. 26, 2017), https://www.nytimes.com/2017/04/26/books/famous-misquotations.html.

[141] *See, e.g.*, Andrew Burt, *The AI Transparency Paradox*, HARV. BUS. REV. (Dec. 13, 2019), https://hbr.org/2019/12/the-ai-transparency-paradox; Bram Vaassen, *AI, Opacity, and Personal Autonomy*, PHIL. & TECH., Sept. 22, 2022, Article No. 88, at 1, https://doi.org/10.1007/s13347-022-00577-5 [https://perma.cc/R9QV-UUNS].

[142] PASQUALE, *supra* note 12, at 4–6.

[143] *See* ALLEN, *supra* note 12, at 59–62 (discussing problems connected with the opacity of the algorithms and models behind financial technology).

mystery to its human overseers.[144] This inability of market participants and regulators to fully comprehend the decision-making processes of AI can lead to significant principal–agent problems and serious systemic implications.[145]

The opacity of AI algorithms, often beyond human interpretability, renders it quite challenging to predict, diagnose, and rectify issues promptly or to precisely address issues after the fact.[146] This increased opacity often obscures the identification and correction of systemic vulnerabilities and market misconduct.[147] AI's proficiency in pattern recognition and predictive analytics, while beneficial for market efficiency, can also lead to new forms of market manipulation and financial misconduct.[148] Sophisticated AI systems could autonomously discover and exploit market vulnerabilities in violation of legal rules or ethical norms before human operators or regulators can detect and mitigate such misdeeds.[149] In isolation, such exploitation would lead to ill-gotten gains and losses for market participants. On a larger scale, such exploitation could jeopardize the stability and integrity of entire markets in the long run as participants withdraw from the market for fear of exploitation.[150] A perhaps instructive analogous development is happening in the world of online poker, where the secret, illicit use of AI bots posing as human players has pushed many players to play with less significant sums or withdraw from playing completely in many forums.[151] Simply put, rigged games draw fewer players.

Compounding the opacity risks is the fact that the wondrous powers and elegant, precise outputs of AI systems can often lull humans into believing that such smart machines are infallible when it comes to matters of money and

144 *See, e.g.*, Shannon Vallor & George A. Bekey, *Artificial Intelligence and the Ethics of Self-Learning Robots*, *in* ROBOT ETHICS 2.0, at 338, 338 (Patrick Lin et al. eds., 2017) ("[S]elf-learning systems behave in ways that cannot always be anticipated or fully understood, even by their programmers."); *cf.* MEREDITH BROUSSARD, ARTIFICIAL UNINTELLIGENCE: HOW COMPUTERS MISUNDERSTAND THE WORLD 177–78 (2019) (calling for more "human-centered design" in AI technologies to better safeguard against harmful consequences).

145 *See* BOSTROM, *supra* note 12, at 127–30 (discussing the agency problems associated with AI); *cf.* Fletcher & Le, *supra* note 25, at 301.

146 *See* Fletcher & Le, *supra* note 25, at 301; ALLEN, *supra* note 12, at 59–62.

147 *See* ALLEN, *supra* note 12, at 59–62.

148 *See, e.g.*, Kristin Johnson, Frank Pasquale & Jennifer Chapman, *Artificial Intelligence, Machine Learning, and Bias in Finance: Toward Responsible Innovation*, 88 FORDHAM L. REV. 499, 510–12 (2019) (discussing how marginalized individuals will be vulnerable to exploitation and discrimination by fintech firms as sophisticated learning algorithms evolve).

149 *See, e.g.*, *id.* at 507 (suggesting that machines can detect and exploit market vulnerabilities without regard for market and legal norms); MELANIE MITCHELL, ARTIFICIAL INTELLIGENCE: A GUIDE FOR THINKING HUMANS 120–21 (2019).

150 *See* PATTERSON, *supra* note 12, at 9–10.

151 *See* Kit Chellel, *The Russian Bot Army That Conquered Online Poker*, BLOOMBERG BUSINESSWEEK (Sept. 20, 2024), https://www.bloomberg.com/features/2024-poker-bots-artificial-intelligence-russia.

markets.[152] Part of the great appeal of artificial financial intelligence is rooted in the fact that old-fashioned, natural intelligence can seem so lacking in our financial markets at times.[153] As such, while many in finance understandably admire the power, precision, and speed of AI-driven machines, they often fail to fully appreciate the limitations of AI to fully code and decode the complexities of financial markets where complicated human actors remain key participants.[154] The elegantly specific outputs of AI systems are often predictions based on data, not perfect answers or perfect information.[155] Ultimately, financial risk, uncertainty, and the "animal spirits" of humans in markets cannot be perfectly decoded, modeled, or mitigated because human behavior can be irrational and unpredictable, even for intelligent computer codes and advanced mathematical models.[156] In fact, a key contributing factor to the depth and velocity of the financial crisis of 2008 was that many of the predominant models of the time failed to properly account for the serious risks in the American housing market.[157]

As AI becomes more sophisticated, powerful, and anthropomorphic, its abilities to seduce humans to trust and rely on its seemingly neutral, objective,

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152 *See* ROBERT J. SHILLER, FINANCE AND THE GOOD SOCIETY 132–33 (2012) (cautioning against overreliance on elegant financial models); David H. Bailey, Jonathan M. Borwein, Marcos López de Prado & Qiji Jim Zhu, *Pseudo-Mathematics and Financial Charlatanism: The Effects of Backtest Overfitting on Out-of-Sample Performance*, 61 NOTICES AM. MATHEMATICAL SOC'Y 458, 458–59 (2014); EMANUEL DERMAN, MODELS. BEHAVING. BADLY. 143–87 (2011).

153 *See* ROBERT Z. ALIBER, CHARLES P. KINDLEBERGER & ROBERT N. MCCAULEY, MANIAS, PANICS, AND CRASHES: A HISTORY OF FINANCIAL CRISES 37–41 (8th ed. 2023) (discussing manias and other unsound financial behavior).

154 *See, e.g.*, RUSSELL, *supra* note 12, at 25–27 (explaining how AI often operates on the erroneous assumption that humans are largely rational decision-makers); Henry T. C. Hu, *Too Complex to Depict? Innovation, "Pure Information," and the SEC Disclosure Paradigm*, 90 TEX. L. REV. 1601, 1608–12 (2012) (discussing the difficulties of properly depicting certain financial market complexities). *But see* JAMES OWEN WEATHERALL, THE PHYSICS OF WALL STREET 36–39 (2013) (opining on the potential of improving financial models to better manage financial risk).

155 AGRAWAL, GANS & GOLDFARB, *supra* note 12, at 24–30.

156 *See* JEROME FRANK, LAW AND THE MODERN MIND 129 (New Brunswick 2009) (1930) ("The acts of human beings are not identical mathematical entities; the individual cannot be eliminated as, in algebraic equations, equal quantities on the two sides can be cancelled."); GEORGE A. AKERLOF & ROBERT J. SHILLER, ANIMAL SPIRITS: HOW HUMAN PSYCHOLOGY DRIVES THE ECONOMY, AND WHY IT MATTERS FOR GLOBAL CAPITALISM 167–70 (9th prtg. 2010) (2009) (discussing the complexities of human psychology and behavior in markets). *See generally* FRANK H. KNIGHT, RISK, UNCERTAINTY AND PROFIT 347 (1921).

157 *See, e.g.*, Paul Krugman, *How Did Economists Get It So Wrong?*, N.Y. TIMES MAG. (Sept. 2, 2009), https://www.nytimes.com/2009/09/06/magazine/06Economic-t.html [https://perma.cc/U67R-CWT4]; *see also* ANTHONY SAUNDERS & LINDA ALLEN, CREDIT RISK MEASUREMENT IN AND OUT OF THE FINANCIAL CRISIS 31 (3d ed. 2010); Amir E. Khandani & Andrew W. Lo, *What Happened to the Quants in August 2007?*, J. INV. MGMT., 2007, at 5, 5–9.

and precise outputs despite its data biases will only grow stronger, much in the same way that we have grown to believe the answers from Google, Alexa, and Siri as using them has become more normal.[158] At the same time, the opacity of AI-powered machines will only make it more difficult for us to explain or understand AI's processes and rationale, even in the post-mortem of a financial crisis where misinformation and misconduct are often contributing factors.[159]

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AI will undoubtedly bring many benefits to the financial system, but it will also bring serious systemic risks. As AI-driven financial activities proliferate, market participants, financial regulators, and other key stakeholders must grow more watchful of systemic risks related to speed and opacity and not blindly subscribe to the hype of AI as a miracle cure-all "snake oil" of bygone eras.[160] The high speeds and dense opacity of financial AI can ultimately undermine investor trust and confidence in the market's integrity and deter investment, thereby leading to market inefficiencies and higher costs of capital. Ultimately, AI has not just created new threats and risks to the marketplace but perhaps an entirely new marketplace altogether.

B. *Geopolitical Threats*

The proliferation of AI in finance introduces a new dimension of geopolitical risk and conflict as adversarial nation-states and nonstate actors leverage AI to attack rivals and adversaries.[161] Rather than engaging in costly and protracted direct military conflicts with uncertain outcomes, nation-states

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[158] *See* Harry Surden, *Artificial Intelligence and Law: An Overview*, 35 GA. ST. U. L. REV. 1305, 1336 (2019); ALLEN, *supra* note 12, at 59; Emily M. Bender, Timnit Gebru, Angelina McMillan-Major & Shmargaret Shmitchell, *On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?*, ASS'N FOR COMPUTING MACH., CONF. ON FAIRNESS, ACCOUNTABILITY, & TRANSPARENCY (2021), at 610–11 (discussing the underappreciated potential harms like "subtle biases" associated with AI applications trained with large language models); Thomas M. Brill, Laura Munoz & Richard J. Miller, *Siri, Alexa, and Other Digital Assistants: A Study of Customer Satisfaction with Artificial Intelligence Applications*, 35 J. MKTG. MGMT. 1401, 1401–10 (2019) (studying how customers grow to trust digital assistants).

[159] *See* ALLEN, *supra* note 12, at 61–62.

[160] *See* ARVIND NARAYANAN & SAYASH KAPOOR, AI SNAKE OIL: WHAT ARTIFICIAL INTELLIGENCE CAN DO, WHAT IT CAN'T, AND HOW TO TELL THE DIFFERENCE 22–26 (2024) (discussing the various harms of overrelying on the hype surrounding early iterations of AI).

[161] *See, e.g.*, HENRY A. KISSINGER, ERIC SCHMIDT & DANIEL HUTTENLOCHER, THE AGE OF AI: AND OUR HUMAN FUTURE 135–42 (2022); Ryan Heath & Megan Morrone, *What U.S.–China Cooperation on AI Could Look Like*, AXIOS (Feb. 2, 2024), https://www.axios.com/2024/02/02/us-china-generative-ai [https://perma.cc/7T6H-GWWW]; KAI-FU LEE, AI SUPERPOWERS: CHINA, SILICON VALLEY, AND THE NEW WORLD ORDER 82–86 (2018).

increasingly resort to economic and business warfare.[162] The United States, including its institutions and businesses, is "particularly vulnerable" to business warfare given its dual role as a dominant global military and economic power.[163] Because defeating the United States in a direct military battle is unlikely and military engagement alone is highly costly, adversaries often choose to wage war through cyber and economic attacks.[164] The integration of AI systems into many aspects of our daily lives and our financial institutions creates many more critical potential points of vulnerability for adversaries to exploit and hack as expressions of national power and defense.[165]

Even before the most recent advances in financial AI, bad actors were leveraging the technology underlying our financial system to attack us and our firms.[166] In 2016, then-Federal Reserve Chair Janet Yellen, in Congressional testimony, explicitly identified cyberattacks on the financial system as "one of the most significant risks our country faces."[167] In 2024, the U.S. Treasury Department, led by Janet Yellen as Treasury Secretary, disclosed a "major breach" via a hack by a Chinese intelligence agency.[168] Geopolitical bad actors and adversaries equipped with relatively inexpensive and highly accessible AI tools can readily engineer sophisticated attacks that exploit weaknesses in

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[162] Tom C.W. Lin, *Business Warfare*, 63 B.C. L. REV. 1, 3–4 (2023).

[163] *Id.* at 9.

[164] *Id.*

[165] *See, e.g.*, NAT'L SEC. COMM'N ON A.I., *supra* note 15, at 24–25 ("AI is the fulcrum of a broader technology competition in the world. AI will be leveraged to advance all dimensions of national power, from healthcare to food production to environmental sustainability."); David E. Sanger & Julian E. Barnes, *China's Hacking Reached Deep into U.S. Telecoms*, N.Y. TIMES (Nov. 21, 2024). https://www.nytimes.com/2024/11/21/us/politics/china-hacking-telecommunications.html; Mark Mazzetti & Edward Wong, *Inside U.S. Efforts to Untangle an A.I. Giant's Ties to China*, N.Y. TIMES (Nov. 27, 2023), https://www.nytimes.com/2023/11/27/us/politics/ai-us-uae-china-security-g42.html.

[166] *See, e.g.*, FIREEYE, APT28: A WINDOW INTO RUSSIA'S CYBER ESPIONAGE OPERATIONS? 3–6 (2014), https://services.google.com/fh/files/misc/apt28-window-russia-cyber-espionage-operations.pdf [https://perma.cc/BF3C-PC3T]; BARRY VENGERIK, KRISTEN DENNESEN, JORDAN BERRY & JONATHAN WROLSTAD, FIREEYE, HACKING THE STREET? FIN4 LIKELY PLAYING THE MARKET 3 (2014), https://www.mandiant.com/sites/default/files/2021-09/rpt-fin4.pdf [https://perma.cc/KK44-W9UV]; Nicole Perlroth & Quentin Hardy, *Bank Hacking Was the Work of Iranians, Officials Say*, N.Y. TIMES (Jan. 8, 2013), https://www.nytimes.com/2013/01/09/technology/online-banking-attacks-were-work-of-iran-us-officials-say.html; Michael Riley, *How Russian Hackers Stole the Nasdaq*, BLOOMBERG BUSINESSWEEK (July 21, 2014), https://www.bloomberg.com/news/articles/2014-07-17/how-russian-hackers-stole-the-nasdaq.

[167] Chiara Albanese, Daniele Lepido & Giles Turner, *'Anonymous' Joins Hacker Army Targeting Central Banks for Cash*, BLOOMBERG L. (Mar. 17, 2017), https://www.bloomberg.com/news/articles/2017-03-17/-anonymous-joins-hacker-army-targeting-central-banks-for-cash.

[168] Ana Swanson & David E. Sanger, *China Hacked Treasury Dept. in 'Major' Breach, U.S. Says*, N.Y. TIMES (Dec. 30, 2024), https://www.nytimes.com/2024/12/30/us/politics/china-hack-treasury.html.

financial AI systems, whether through market manipulation, financial misinformation, or fraudulent transactions—all intended to harm an adversary.[169]

Recent geopolitical conflicts involving Russia and Ukraine, Israel and Hamas, as well as China and the United States foreshadow the looming threat of AI being used as a weapon against rival state actors in a myriad of ways for various strategic ends.[170] Rogue states and terrorist organizations can use AI to spread misinformation and sow discord to damage rival economies, procure illicit gains, or hurt the political prospects of foreign leaders.[171] Russia is believed to have interfered in both the 2016 and 2020 U.S. presidential elections through fake social media accounts that disseminated misinformation aimed at influencing American voters.[172] And in the 2024 U.S. presidential election cycle, Russia, China, and Iran all engaged in misinformation campaigns using AI and other means to influence the electorate.[173] It is not hard to imagine a foreign adversary using AI-driven misinformation and market misconduct to destabilize our financial markets and economy in advance of a presidential election to hurt one candidate and help another one win.[174] Furthermore, as businesses engage more actively and openly in political issues and processes,

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[169] *See* NAT'L SEC. COMM'N ON A.I., *supra* note 15, at 45–46; Lin, *supra* note 128, at 538–39.

[170] *See* NAT'L SEC. COMM'N ON A.I., *supra* note 15, at 7 ("AI is deepening the threat posed by cyber attacks and disinformation campaigns that Russia, China, and others are using to infiltrate our society, steal our data, and interfere in our democracy."); *see also* Vera Bergengruen, *How Tech Giants Turned Ukraine into an AI War Lab*, TIME (Feb. 8, 2024), https://time.com/6691662/ai-ukraine-war-palantir [https://perma.cc/9K2C-S3VV] (discussing Ukraine's use of AI to mine battlefield data quickly).

[171] *See* NAT'L SEC. COMM'N ON A.I., *supra* note 15, at 45–46.

[172] *See* Sheera Frenkel & Julian E. Barnes, *Russians Again Targeting Americans with Disinformation, Facebook and Twitter Say*, N.Y. TIMES, https://www.nytimes.com/2020/09/01/technology/facebook-russia-disinformation-election.html (Sept. 22, 2020); Greg Myre, *Intelligence Report: Russia Tried to Help Trump in 2020 Election*, NPR (Mar. 16, 2021), https://www.npr.org/2021/03/16/977958302/intelligence-report-russia-tried-to-help-trump-in-2020-election [https://perma.cc/8SAU-UAWC]; Julian E. Barnes, *Russian Interference in 2020 Included Influencing Trump Associates, Report Says*, N.Y. TIMES, https://www.nytimes.com/2021/03/16/us/politics/election-interference-russia-2020-assessment.html (May 27, 2021).

[173] David E. Sanger & Julian E. Barnes, *Crunchtime for Election Interference: October Is the Month of Mischief*, N.Y. TIMES (Sept. 26, 2024), https://www.nytimes.com/2024/09/26/us/politics/october-election-interference.html.

[174] *See, e.g.*, Daniel I. Weiner & Lawrence Norden, *Regulating AI Deepfakes and Synthetic Media in the Political Arena*, BRENNAN CTR. FOR JUST. (Dec. 5, 2023), https://www.brennancenter.org/our-work/research-reports/regulating-ai-deepfakes-and-synthetic-media-political-arena [https://perma.cc/9H6D-SKVS].

the odds of private firms antagonizing and then being targeted by a foreign adversary also increase over time.[175]

The inherent complexity and opacity of AI algorithms, especially those based on deep self-learning, exacerbate the deleterious impact of these geopolitical threats.[176] The black-box nature of many AI systems can obscure their inner workings, making it challenging to predict or counteract the myriad strategies employed by adversarial actors.[177] Furthermore, autonomous, AI-driven financial systems increase the speed and scale at which an acute attack on one firm or industry can quickly transform into a catastrophic systemic threat.[178] For instance, an AI-powered misinformation attack from a rogue state against the market for the stock of Apple, one of the most valuable companies in the world, could quickly destabilize the entire system as various index funds, mutual funds, debt instruments, and credit derivatives linked to Apple equity would all react to the attack in a volatile manner.

In sum, the rise of AI as an essential tool in finance escalates the probability and impact of geopolitical threats to our financial markets as these tools are repurposed as weapons in geopolitical conflicts.[179] AI-powered financial systems and operations expose many critical points for manipulation, misconduct, and malice for adversaries to exploit against their geopolitical rivals—particularly the United States, the largest economy and most important player on the global stage.[180]

C. *Normal Financial Accidents*

The proliferation of AI in finance increases the likelihood of catastrophic financial accidents.[181] Charles Perrow, in his seminal study on technological risks titled *Normal Accidents: Living with High-Risk Technologies*, argued that intricate, complex technology systems are inherently susceptible to calamities and mishaps.[182] As financial AI grows more prevalent, "normal financial

[175] *See* Tom C.W. Lin, *The New Corporate Political Governance*, 65 B.C. L. REV. 833, 842–45 (2024) (discussing the rising convergence of corporate and political governance as businesses engage more actively on sociopolitical issues).

[176] *See, e.g.*, Johnson, Pasquale & Chapman, *supra* note 148, at 507 (discussing problems and risks that can arise because "much of contemporary AI is either opaque or so complex").

[177] *See* PASQUALE, *supra* note 12, at 4–6; Andrew D. Selbst & Solon Barocas, *The Intuitive Appeal of Explainable Machines*, 87 FORDHAM L. REV. 1085, 1089–96 (2018).

[178] *See, e.g.*, Kirilenko & Lo, *supra* note 12, at 60; Lin, *supra* note 23, at 711–14.

[179] *See* LEE, *supra* note 161, at 82–86; SMITH & BROWNE, *supra* note 14, at 69–76.

[180] *See* NAT'L SEC. COMM'N ON A.I., *supra* note 15, at 7; Robin Sidel, *Banks Battle Staffers' Vulnerability to Hacks*, WALL ST. J. (Dec. 20, 2015), http://www.wsj.com/articles/the-weakest-link-in-banks-fight-against-hackers-1450607401.

[181] SUBCOMM. ON EMERGING & EVOLVING TECHS., *supra* note 22, at 47.

[182] *See* CHARLES PERROW, NORMAL ACCIDENTS: LIVING WITH HIGH-RISK TECHNOLOGIES 4–5 (Princeton Univ. Press 1999) (1984).

accidents" are likely to become more common within the financial system.[183] Within the last twenty years, both the New York Stock Exchange and the Nasdaq, two of the most prominent and technologically advanced stock exchanges in the world, suffered unprecedented malfunctions that halted billions of dollars' worth of trading for hours during otherwise normal trading sessions.[184]

The deployment of AI in critical financial operations through tools like trading algorithms, risk-management models, and predictive analytics adds yet another layer of technological complexity and interdependence to an already-high-tech and interconnected financial marketplace.[185] The rapid capabilities and powers of AI systems mean that when accidents occur, they have the potential to escalate swiftly, outpacing human oversight and intervention. Given the highly intermediated nature of modern finance and the use of similar and interdependent AI programs by many financial firms, even minor errors can propagate and interact in unforeseen ways, leading to significant failures.[186] A small glitch in the AI systems of one or a few firms—regardless of their size or market value—can create destabilizing ripples, leading to volatile cycles and catastrophic cascades for the entire financial system.[187] As economists Andrei Kirilenko and Andrew Lo astutely wrote, "whatever can go wrong will go wrong faster and bigger when computers are involved."[188]

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[183] *See* Marc Schneiberg & Tim Bartley, *Regulating or Redesigning Finance? Market Architectures, Normal Accidents, and Dilemmas of Regulatory Reform*, 30A RSCH. SOCIO. ORGS. 281, 284–89 (2010).

[184] *See* E. S. Browning & Scott Patterson, *Market Size + Complex Systems = More Glitches*, WALL ST. J. (Aug. 22, 2013), https://www.wsj.com/articles/SB100014241278873239806045790293420015341 48; Nathaniel Popper, *Pricing Problem Suspends Nasdaq for Three Hours*, N.Y. TIMES (Aug. 22, 2013), https://dealbook.nytimes.com/2013/08/22/nasdaq-market-halts-trading [https://perma.cc/UK97-VVCR]; Nathaniel Popper, *The Stock Market Bell Rings, Computers Fail, Wall Street Cringes*, N.Y. TIMES (July 8, 2015), https://www.nytimes.com/2015/07/09/business/dealbook/new-york-stock-exchange-suspends-trading.html.

[185] *See* AGRAWAL, GANS & GOLDFARB, *supra* note 12, at 29–30 (highlighting the limitations of AI predictions and forecasts); Markus K. Brunnermeier, *Deciphering the Liquidity and Credit Crunch 2007–2008*, J. ECON. PERSPS., Winter 2009, at 77, 96–97 (explaining how "an interwoven network of financial obligations" serves as a key foundation of the financial system); Tom C.W. Lin, *Infinite Financial Intermediation*, 50 WAKE FOREST L. REV. 643, 661 (2015) (discussing the highly intermediated nature of modern finance).

[186] *See* Lin, *supra* note 128, at 542 (discussing how AI can cause "actions, errors, and failings [to] trigger destabilizing ripples across the financial system because of the interconnectivity of firms, regardless of their value or size").

[187] *See* PATTERSON, *supra* note 12, at 9–10 (explaining how financial technology can induce "a vicious self-reinforcing feedback loop"); Surowiecki, *supra* note 64; Chris Brummer & Yesha Yadav, *Fintech and the Innovation Trilemma*, 107 GEO. L.J. 235, 237–38 (2019) (discussing market disruptors and the need for regulatory response); Steven L. Schwarcz, *Systemic Risk*, 97 GEO. L.J. 193, 199–200 (2008) (discussing the systemic risks associated with financial intermediation).

[188] Kirilenko & Lo, *supra* note 12, at 52.

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The rising integration of AI into financial markets increases the dangers of new systemic risks, geopolitical threats, and major financial accidents.[189] While there are many unfolding gains and innovations to be derived from financial AI, such benefits can only be sustainable and fully harvested if the marketplace is vigilant of the growing structural dangers and challenges posed by this new financial technology.[190]

IV. Regulatory and Policy Challenges

Regulating the proliferation of AI in finance will be one of the most significant regulatory and policy challenges in the years to come.[191] The rise of financial AI raises fundamental issues at the core of preventing and prosecuting financial misconduct, as well as protecting investors, innovation, and the integrity of the market.[192]

A. *Limitations of Law, Detection, and Resources*

Regulating misconduct connected to AI-powered financial technologies reveals critical gaps and limitations in the current legal and regulatory frameworks related to foundational issues of law, enforcement, and resources.[193]

1. *Limitations of Law*

Many existing laws and regulations, conceived and crafted in analog and pre-AI financial eras, are not well designed to address financial misconduct propagated by autonomous, high-tech systems during our AI-driven financial

[189] *See, e.g.*, RUSSELL, *supra* note 12, at 103–10 (describing various ways to misuse AI).

[190] *See* PASQUALE, *supra* note 12, at 185–88 (outlining a policymaking approach to better balance the innovative benefits of AI system and its deleterious effects); Fletcher & Le, *supra* note 25, at 322.

[191] *See* Brummer & Yadav, *supra* note 187, at 237–44, 306.

[192] *Id.* at 244 (discussing the "trilemma" of regulating financial technology with the objectives of "(i) market integrity; (ii) rules simplicity; and (iii) financial innovation").

[193] *See, e.g.*, Fletcher & Le, *supra* note 25, at 292 ("AI does not easily fit the categorizations within the existing legal framework, resulting in regulatory gaps, reactive rather than proactive regulation, or ill-fitting frameworks that exacerbate risks and cramp innovation."); Tom C.W. Lin, *The New Financial Industry*, 65 ALA. L. REV. 567, 593–95 (2014) (discussing various challenges to financial regulation in a high-tech marketplace).

age.[194] These systems, characterized by their ability to learn, adapt, and make decisions algorithmically and autonomously without human intervention, pose unique challenges that existing laws are ill-equipped to address directly or elegantly.[195]

Traditional legal approaches to financial misconduct—like those in corporate law and securities law, which rely heavily on attributing actions and imposing liability on human actors with discernible bad intentions—are rendered ineffectual when misconduct is a result of autonomous AI algorithms.[196] Historically, holding accountable perpetrators of market manipulation schemes and financial misconduct often requires culpable intent from a natural person as a key underlying element.[197] The inherent opacity of complex AI systems, their "black box" decision-making processes, and the absence of clear legal personhood and scienter complicate the ascription of liability and accountability.[198]

Moreover, the rapid evolution of AI technologies outpaces the often-glacial legislative and rulemaking processes, leading to regulatory lags that exacerbate the inadequacy of existing laws when new principles and definitions are necessary.[199] The agility of AI systems in adapting to new data and market

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[194] *See* MARKHAM, *supra* note 12, at 390–91; Gregory Scopino, *Do Automated Trading Systems Dream of Manipulating the Price of Futures Contracts? Policing Markets for Improper Trading Practices by Algorithmic Robots*, 67 FLA. L. REV. 221, 222 (2015) ("Today, federal regulators are faced with a very different and yet in some ways similar task: monitoring the actions of artificially-intelligent algorithmic trading robots—frequently referred to as 'algo bots'—in a continuous effort to combat price manipulation and disruptive trading practices in the markets . . . .").

[195] *See* Carsten Gerner-Beuerle, *Algorithmic Trading and the Limits of Securities Regulation*, *in* DIGITAL FINANCE IN EUROPE: LAW, REGULATION, AND GOVERNANCE 109, 112 (2022) (arguing that preexisting regulatory frameworks based on disclosure are insufficient to address new market challenges); Arbel, Tokson & Lin, *supra* note 45, at 552 ("[T]he doctrinal apparatus developed to regulate existing technologies is ill-equipped to deal with the unique risk of highly capable AI systems.").

[196] *See* Yesha Yadav, *The Failure of Liability in Modern Markets*, 102 VA. L. REV. 1031, 1034–39 (2016); John Lightbourne, *Algorithms & Fiduciaries: Existing and Proposed Regulatory Approaches to Artificially Intelligent Financial Planners*, 67 DUKE L.J. 651, 652–53 (2017); Mihailis E. Diamantis, *Employed Algorithms: A Labor Model of Corporate Liability for AI*, 72 DUKE L.J. 797, 801 (2023) ("[C]urrent law falls short when corporate algorithms, rather than employees, cause harm.").

[197] MARKHAM, *supra* note 12, at 375.

[198] *See, e.g.*, Scopino, *supra* note 194, at 250 ("[S]cienter—or a culpable mental state—is a required element of the majority of civil claims involving manipulation, abusive market practices, or financial fraud. Only humans and business entities are considered 'persons' for purposes of the law. Noticeably, that leaves out computers and software programs . . . ." (footnotes omitted)); Azzutti, *supra* note 44, at 10 (describing the "challenges posed by AI-driven manipulation to achieve effective law enforcement").

[199] *See, e.g.*, Margot E. Kaminski & Jennifer M. Urban, *The Right to Contest AI*, 121 COLUM. L. REV. 1957, 1960–62 (2021) (examining the unmet need for a legal "right to

conditions often leads to emergent behaviors that existing risk assessment models and compliance checklists cannot foresee or contain in a timely manner.[200] This technological dynamism, while beneficial for innovation and efficiency in financial services, creates systemic risks that traditional regulatory tools, such as periodic reporting and post-hoc enforcement actions, are often too slow to mitigate.[201] While regulators and policymakers have valiantly attempted to retrofit traditional legal rules of financial regulation to the new financial marketplace in the past, it will be difficult to adapt rules designed to govern natural persons with states of mind to autonomous, AI-driven systems.[202]

2. *Limitations of Detection*

Even placing aside the limitations of current law in terms of enforcement, detection poses significantly greater challenges with AI-driven financial misinformation and misconduct as compared to human-driven misbehavior.[203] It is incredibly hard to detect misconduct and misinformation originating from high-speed AI-powered systems in a timely manner when transactions occur at velocities measured in fractions of a second in a data-rich marketplace; moreover, AI-powered systems may self-learn to better evade detection and liability.[204] The ability of firms to harness vast datasets and computational power allows for the constant refinement and optimization of AI models, often in ways that are opaque and not readily understandable even to their creators, let alone external regulators in real time.[205]

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contest AI" as part of due process protections for individuals in the United States); Verstein, *supra* note 52, at 272 (discussing the legal challenges of defining manipulation in a dynamic marketplace).

200 *See, e.g.*, Scopino, *supra* note 194, at 233 ("[C]auses of action [requiring scienter] would be ineffective in circumstances where computerized trading bots, without specific human direction, engaged in disruptive trading conduct while continuously modifying their own algorithms . . . .").

201 *See* Lin, *supra* note 40, at 1303 ("[U]ntil new precedents, principles, and rules are firmly established, there will be significant enforcement challenges for regulators as they combat the new methods of market manipulation.").

202 *See* LARRY DOWNES, THE LAWS OF DISRUPTION: HARNESSING THE NEW FORCES THAT GOVERN LIFE AND BUSINESS IN THE DIGITAL AGE 18–19 (2009) (discussing difficulties in adapting old legal rules to new technology); *see also* SAMIR CHOPRA & LAURENCE F. WHITE, A LEGAL THEORY FOR AUTONOMOUS ARTIFICIAL AGENTS 153–63 (2011); Lawrence B. Solum, *Legal Personhood for Artificial Intelligences*, 70 N.C. L. REV. 1231, 1231–33 (1992).

203 *See* Lin, *supra* note 40, at 1296 ("The emerging modes of cybernetic market manipulation are particularly challenging for resource-constrained regulators because they are incredibly difficult to detect due to the accelerated speed, data deluge, and balkanization of the marketplace.").

204 *See* Azzutti, Ringe & Stiehl, *supra* note 87, at 116 ("AI may subvert established market conduct rules providing for detection, liability attribution, and other enforcement mechanisms.").

205 PASQUALE, *supra* note 12, at 4–14.

Exacerbating the challenges of detecting misinformation, manipulation, and misconduct is the fact that the marketplace is awash in data even before the latest growth in data analytics.[206] Market-moving information can manifest in various forms and sources, not only financial or firm-based information but also social media data, consumer metadata, and other data traditionally unrelated to finance or business.[207] Once misinformation enters the financial system, it can spread rapidly and evolve as market participants—through machines and humans—react to it.[208] This dynamic process has shortened the average holding period for securities in the United States to mere seconds as bots constantly reallocate capital in response to new informational stimuli.[209] One estimate in 2019 stated that on any given day, the world generates over 2.5 quintillion bytes or 1 billion gigabytes of data, including 500 million tweets and 294 billion emails, with these figures only increasing each year.[210] Attempting to timely detect misinformation, misconduct, and manipulative schemes in this modern data environment is less like finding a needle in a haystack and more like finding a particular grain of sand in an exponentially expanding beach.

Furthermore, even when detected, attributing financial misconduct or market-moving misinformation to the responsible party in a timely manner proves incredibly challenging when the information travels through cyberspace and global informational networks.[211] While regulators can study audit trails after the fact, attribution nevertheless has remained a persistent problem with bad acts conducted in cyberspace and private electronic networks, entailing complex technical, legal, policy, and geopolitical considerations.[212] Without

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[206] *See* ROBERT S. KRICHEFF, DATA ANALYTICS FOR CORPORATE DEBT MARKETS: USING DATA FOR INVESTING, TRADING, CAPITAL MARKETS, AND PORTFOLIO MANAGEMENT 55–60 (2014) (discussing the value of data mining and the various data-analyzing techniques prior to the rise of AI); FRANK J. OHLHORST, BIG DATA ANALYTICS: TURNING BIG DATA INTO BIG MONEY 37–46 (2012); Lin, *supra* note 40, at 1297–300.

[207] *See* OHLHORST, *supra* note 206, at 37–46.

[208] *See* Lin, *supra* note 40, at 1297 ("[T]he increasing influence of digital data and information has made detecting the new methods of market manipulation much more challenging for regulators.").

[209] *See* PATTERSON, *supra* note 12, at 46 ("At the end of World War II, the average holding period for a stock was four years. By 2000, it was eight months. By 2008, it was two months. And by 2011 it was *twenty-two seconds* . . . .").

[210] Jeff Desjardins, *How Much Data Is Generated Each Day?*, WORLD ECON. F. (Apr. 17, 2019), https://www.weforum.org/agenda/2019/04/how-much-data-is-generated-each-day-cf4bddf29f [https://perma.cc/Q7ML-TGMR].

[211] *See, e.g.*, Jens David Ohlin, *Cyber Causation*, *in* CYBERWAR: LAW AND ETHICS FOR VIRTUAL CONFLICTS 37–44 (Jens David Ohlin et al. eds., 2015); Michael N. Schmitt & Liis Vihul, *Proxy Wars in Cyberspace: The Evolving International Law of Attribution*, FLETCHER SEC. REV., Spring 2014, at 55, 55.

[212] *See* Eichensehr, *supra* note 12, at 524–27 (outlining a myriad of complicated legal and political issues in connection with cyberattack attribution); JOHN S. DAVIS II ET AL., STATELESS ATTRIBUTION: TOWARD INTERNATIONAL ACCOUNTABILITY IN CYBERSPACE 9–16 (2017).

proper attribution, enforcement efforts to hold bad actors accountable for financial misconduct are not possible.[213] The challenges surrounding attribution are further exacerbated by the opaque black-box nature of many AI technologies and the advanced capabilities of bad actors to cover their digital tracks from proper detection and attribution.[214]

3. *Limitations of Resources*

The resource asymmetries between public regulators and private industry pose significant challenges for effectively regulating AI-driven financial misconduct, particularly within the rapidly evolving landscape of financial technologies.[215] The capital and expertise required to keep up with advancements in AI are incredibly expensive and competitive.[216] The public sector often lacks the resources to acquire the latest, most advanced technologies or hire the most talented and sought-after tech experts.[217]

In contrast, private entities, especially large corporations in the financial sector, command vast resources and cutting-edge technological capabilities, enabling them to develop and deploy sophisticated AI systems.[218] These entities

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[213] *See* Tom C.W. Lin, *Financial Weapons of War*, 100 MINN. L. REV. 1377, 1425 (2016) ("[T]he issue of accountability is predicated on the notion that wrongdoers can be properly identified for their misdeeds."); Nicholas Tsagourias, *Cyber Attacks, Self-Defence and the Problem of Attribution*, 17 J. CONFLICT & SEC. L. 229, 233 (2012) ("What is critical, then, is not only to trace back the attack to its source, for example to a computer, but to identify the person who operated the computer, and more importantly to identify the real 'mastermind' behind the attack . . . .").

[214] *See, e.g.*, PASQUALE, *supra* note 12, at 4–6.

[215] *See* Gianclaudio Malgieri & Frank Pasquale, *Licensing High-Risk Artificial Intelligence: Toward Ex Ante Justification for a Disruptive Technology*, COMPUT. L. & SEC. REV., Apr. 2024, Article No. 105899, at 2.

[216] *See id.* at 1–2.

[217] Even prior to the latest advances in AI and financial technology, lack of sufficient funding had been a perennial problem for financial regulators. *See, e.g.*, *Testimony on Budget and Management of the U.S. Securities and Exchange Commission: Hearing Before the H. Comm. on Fin. Servs. Subcomm. on Cap. Mkts., Ins. & Gov't-Sponsored Enters.*, 112th Cong. (2011) (statement of Robert Khuzami, Dir., Div. of Enf't et al.), http://www.sec.gov/news/testimony/2011/ts031011directors.htm [https://perma.cc/E32S-JDC4] ("Over the past decade, the SEC has faced significant challenges in maintaining a staffing level and budget sufficient to carry out its core mission. The SEC experienced three years of frozen or reduced budgets . . . that forced a reduction of 10 percent of the agency's staff. Similarly, the agency's investments in new or enhanced IT systems declined about 50 percent . . . ."); Arthur Levitt Jr., Opinion, *Don't Gut the S.E.C.*, N.Y. TIMES (Aug. 7, 2011), http://www.nytimes.com/2011/08/08/opinion/dont-gut-the-sec.html; James B. Stewart, *As a Watchdog Starves, Wall Street Is Tossed a Bone*, N.Y. TIMES (July 15, 2011), http://www.nytimes.com/2011/07/16/business/budget-cuts-to-sec-reduce-its-effectiveness.html.

[218] *See* Steve Lohr, *Generative A.I.'s Biggest Impact Will Be in Banking and Tech, Report Says*, N.Y. TIMES (Feb. 1, 2024), https://www.nytimes.com/2024/02/01/business/ai-impact-jobs.html.

often heavily invest in AI research and development, harnessing the brightest minds and most cutting-edge technologies to drive innovation and maintain a competitive edge.[219] Public regulatory bodies, on the other hand, are frequently constrained by limited budgets, outdated technological tools, and an overworked, under-resourced staff as they attempt to effectively oversee complex AI systems.[220] Moreover, expertise developed at public agencies often exits via "the revolving door" to private firms offering compensation packages that dwarf government pay.[221] As such, institutional knowledge is difficult to build and sustain for public regulators because of resource asymmetries between public-regulated and private industry.[222] This imbalance not only impedes the regulators' ability to understand the intricacies of AI-driven financial activities but also limits their capacity to develop the sustainable institutional expertise to monitor these activities effectively and enforce regulations proactively.[223]

Furthermore, the pace at which AI and financial technology evolve further exacerbates this resource disparity. AI-driven systems in the financial sector can adapt and learn at velocities that outstrip a regulator's ability to understand, anticipate, and respond to new risks and forms of misconduct.[224] This rapid evolution often results in a regulatory lag, wherein rules and oversight mechanisms become outdated almost as soon as they are implemented.[225] The speed and opacity of AI systems, coupled with the resource asymmetry, present formidable challenges for regulators attempting to detect, understand, and

[219] *See, e.g.*, Erin Griffith & Cade Metz, *A New Area of A.I. Booms, Even Amid the Tech Gloom*, N.Y. TIMES (Jan. 7, 2023), https://www.nytimes.com/2023/01/07/technology/generative-ai-chatgpt-investments.html; Erin Griffith, *Investors Pour $27.1 Billion into A.I. Start-Ups, Defying a Downturn*, N.Y. TIMES (July 3, 2024), https://www.nytimes.com/2024/07/03/technology/ai-startups-funding.html; Michael J. de la Merced & Cade Metz, *OpenAI in Talks for Deal That Would Value Company at $100 Billion*, N.Y. TIMES (Aug. 28, 2024), https://www.nytimes.com/2024/08/28/technology/openai-valuation-funding-deal.html.

[220] *See* Awrey & Judge, *supra* note 12, at 2311.

[221] *See, e.g.*, U.S. GOV'T ACCOUNTABILITY OFF., GAO-11-654, SECURITIES AND EXCHANGE COMMISSION: EXISTING POST-EMPLOYMENT CONTROLS COULD BE FURTHER STRENGTHENED 2 (2011), http://www.gao.gov/new.items/d11654.pdf [https://perma.cc/9C56-SDYZ]; PROJECT ON GOV'T OVERSIGHT, DANGEROUS LIAISONS: REVOLVING DOOR AT SEC CREATES RISK OF REGULATORY CAPTURE 2–10 (2013), https://s3.amazonaws.com/docs.pogo.org/report/2013/20130211-dangerous-liaisons-sec-revolving-door.pdf [https://perma.cc/F7ES-QBJU]; JAMES Q. WILSON & JOHN J. DILULIO JR., AMERICAN GOVERNMENT: INSTITUTIONS AND POLICIES 278 (11th ed. 2008).

[222] *See* PROJECT ON GOV'T OVERSIGHT, *supra* note 221, at 2–10.

[223] *See, e.g.*, Levitt Jr., *supra* note 217; Stewart, *supra* note 217.

[224] Tom Wheeler, *The Three Challenges of AI Regulation*, BROOKINGS (June 15, 2023), https://www.brookings.edu/articles/the-three-challenges-of-ai-regulation [https://perma.cc/RJP4-BG9F].

[225] *See, e.g.*, Satariano & Kang, *supra* note 16 ("A.I. systems are advancing so rapidly and unpredictably that lawmakers and regulators can't keep pace.").

mitigate AI-driven financial misconduct.[226] The resource gap thus not only hinders the effective oversight of AI in finance but also raises concerns about equity, fairness, and power within the financial system. Those with the means to leverage advanced technologies may operate with minimal regulatory oversight or accountability, potentially wielding disproportionate influence and profit at the expense of other market participants.[227] Addressing this imbalance is crucial for maintaining the integrity of financial markets and ensuring that the benefits of AI-driven innovation are realized without compromising market fairness and investor protection.

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Promulgating regulations and policies to respond to financial misconduct powered by AI technologies is hindered by fundamental limitations related to law, enforcement, and resources. These limitations not only inhibit the effective deterrence and remediation of AI-driven misconduct but also fail to provide a workable overarching framework for proactive oversight and compliance that is essential in the fast-paced modern marketplace, where technology, compliance, and market-making are closely integrated with one another.[228]

B. *Politics and Geopolitics*

One of the chief challenges of managing AI in finance is not technological but political.[229] While many proposals and initiatives have been made around the world to regulate AI, actual, meaningful implementation and enforcement remains to be seen.[230] Overcoming the limitations and shortcomings of existing laws and regulations requires implementing new policies—both domestically and internationally, given the global nature of the modern financial

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[226] *See, e.g.*, Awrey & Judge, *supra* note 12, at 2311 ("[G]iven the incredible complexity and dynamism of finance, together with the finite resources of regulators, high information and other costs remain a powerful constraint on who knows what at any point in time.").

[227] *See, e.g.*, PATTERSON, *supra* note 12, at 230 ("The new hierarchy would be all about who owned the most powerful computers, the fastest links between markets, the most sophisticated algorithms—and the inside knowledge of how the market's plumbing was put together.").

[228] *See, e.g.*, Tom C.W. Lin, *Compliance, Technology, and Modern Finance*, 11 BROOK. J. CORP., FIN. & COM. L. 159, 177–79 (2016) (discussing the integration of financial compliance and technology).

[229] *See, e.g.*, Alina Tugend, *Experts on A.I. Agree that It Needs Regulation. That's the Easy Part.*, N.Y. TIMES (Dec. 6, 2023), https://www.nytimes.com/2023/12/06/business/dealbook/artificial-intelligence-regulation.html.

[230] *See, e.g.*, Kolt, *supra* note 129, at 1202–09 (summarizing various regulatory efforts to regulate AI around the world).

marketplace.[231] This is to say nothing of the fact that some politicians in the United States and around the world have used AI-generated images to spread misinformation for their own political ends.[232] Unfortunately, while technologies in AI and the financial sector move expediently, political processes and consensus often lag behind as political leaders leverage the new technology and regulatory void to their own advantage.[233]

Domestically, American politics and political institutions are deeply polarized and dysfunctional.[234] Petty politics and high-powered lobbyists craft the legislative agenda,[235] resulting in much-needed rules and regulations going undrafted or unpassed.[236] For instance, the 118th Congress in 2023 stood out as one of the least productive legislative sessions in decades.[237] The House of Representatives passed just twenty-seven bills that became law despite holding

[231] *See, e.g.*, Bill Whyman, *AI Regulation Is Coming – What Is the Likely Outcome?*, CTR. FOR STRATEGIC & INT'L STUD. (Oct. 10, 2023), https://www.csis.org/blogs/strategic-technologies-blog/ai-regulation-coming-what-likely-outcome [https://perma.cc/Z5W6-YU4E] (discussing the need for global regulation of AI-related issues); Satariano & Kang, *supra* note 16 (reporting on the difficulties of regulating AI domestically and internationally).

[232] *See* Neil Vigdor, *Trump Promotes A.I. Images to Falsely Suggest Taylor Swift Endorsed Him*, N.Y. TIMES (Aug. 19, 2024), https://www.nytimes.com/2024/08/19/us/politics/trump-taylor-swift-ai-images.html; *see also* Tiffany Hsu & Steven Lee Myers, *A.I.'s Use in Elections Sets Off a Scramble for Guardrails*, N.Y. TIMES (June 25, 2023), https://www.nytimes.com/2023/06/25/technology/ai-elections-disinformation-guardrails.html (reporting on the use of AI in connection with spreading political misinformation and disinformation); Tiffany Hsu & Cade Metz, *In Big Election Year, A.I.'s Architects Move Against Its Misuse*, N.Y. TIMES (Feb. 16, 2024), https://www.nytimes.com/2024/02/16/technology/ai-elections-defense.html ("A.I. tools have also created misleading or deceptive portrayals of politicians and political topics in Argentina, Australia, Britain and Canada.").

[233] *See, e.g.*, Awrey & Judge, *supra* note 12, at 2314 (describing financial rulemaking as "a slow, deliberative, and incremental process"); Satariano & Kang, *supra* note 16.

[234] *See* Annie Karni, *House Dysfunction by the Numbers: 724 Votes, Only 27 Laws Enacted*, N.Y. TIMES (Dec. 19, 2023), https://www.nytimes.com/2023/12/19/us/politics/house-republicans-laws-year.html; THOMAS E. MANN & NORMAN J. ORNSTEIN, IT'S EVEN WORSE THAN IT LOOKS: HOW THE AMERICAN CONSTITUTIONAL SYSTEM COLLIDED WITH THE NEW POLITICS OF EXTREMISM X–XIV (2012); EZRA KLEIN, WHY WE'RE POLARIZED 197–200 (2020) (recounting the partisanship at play after Senate Republicans refused to consider President Obama's Supreme Court nominee after the death of Justice Antonin Scalia).

[235] *See* JEFF CONNAUGHTON, THE PAYOFF: WHY WALL STREET ALWAYS WINS 113 (2012).

[236] *See, e.g.*, Roberta S. Karmel, *IOSCO's Response to the Financial Crisis*, 37 J. CORP. L. 849, 853 (2012) ("Where regulated industries have so much power and influence over lawmakers, there is a lack of political will to engage in vigorous regulation even when regulators perceive the dangers of insufficient market place standards.").

[237] Andrew Solender, *Capitol Hill Stunner: 2023 Led to Fewest Laws in Decades*, AXIOS (Dec. 18, 2023), https://www.axios.com/2023/12/19/118-congress-bills-least-unproductive-chart [https://perma.cc/U6MU-APQ9].

724 votes in 2023.[238] In areas such as financial and technology policy, there is often intense lobbying against more government oversight and regulation.[239]

In recent history, almost every major piece of legislation targeting the financial industry or the technology industry has been subjected to incredibly expensive and highly sophisticated efforts to dilute or derail meaningful reforms or new rules.[240] Even legislative responses to major corporate scandals and financial crises, like the Sarbanes-Oxley Act of 2002 and the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010, have been subject to intense lobbying and dilution despite bipartisan support.[241] Not surprisingly, as policymakers and regulators answered calls to regulate AI technologies in 2023, lobbying increased dramatically.[242] One reputable estimate in 2024 found a 185% year-to-year increase in lobbying on AI-related policies by over 450 organizations.[243]

Acknowledging these difficulties of legislative action and domestic politics is not to suggest that complete inaction is inevitable. The President can take executive action to help mitigate the deleterious effects of AI-generated misinformation for national security purposes through executive orders and memos as well as laws like the Protecting Americans from Foreign Adversary Controlled Applications Act and the Foreign Investment Risk Review

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[238] Karni, *supra* note 234.

[239] *See, e.g.*, Rebecca M. Kysar, *The Sun Also Rises: The Political Economy of Sunset Provisions in the Tax Code*, 40 GA. L. REV. 335, 392 (2006) ("Through campaign contributions and lobbyists, these [interest] groups seek legislative votes favorable to their interests from politicians."); Eric Lipton & Ben Protess, *Banks' Lobbyists Help in Drafting Financial Bills*, N.Y. TIMES: DEALBOOK (May 23, 2013), https://dealbook.nytimes.com/2013/05/23/banks-lobbyists-help-in-drafting-financial-bills [https://perma.cc/W9N3-ZB9N]; Emily Birnbaum, *Tech Spent Big on Lobbying Last Year*, POLITICO (Jan. 24, 2022), https://www.politico.com/newsletters/morning-tech/2022/01/24/tech-spent-big-on-lobbying-last-year-00001144 [https://perma.cc/QWY9-XE69]; Inci Sayki, *Big Tech Lobbying on AI Regulation as Industry Races to Harness ChatGPT Popularity*, OPENSECRETS (May 4, 2023), https://www.opensecrets.org/news/2023/05/big-tech-lobbying-on-ai-regulation-as-industry-races-to-harness-chatgpt-popularity [https://perma.cc/5Q5E-Z39J]; Douglas Gillison, *US Bank Lobbyists Ranks Swell to Post-Crisis High Amid Regulatory Pushback*, REUTERS (Feb. 8, 2024), https://www.reuters.com/business/finance/us-bank-lobbyists-ranks-swell-post-crisis-high-amid-regulatory-pushback-2024-02-08.

[240] *See, e.g.*, John C. Coffee Jr., *The Political Economy of Dodd-Frank: Why Financial Reform Tends to Be Frustrated and Systemic Risk Perpetuated*, 97 CORNELL L. REV. 1019, 1020–37 (2012); Roberta Romano, *The Sarbanes-Oxley Act and the Making of Quack Corporate Governance*, 114 YALE L.J. 1521, 1560–68, 1581 (2005); Charles Duhigg, *Silicon Valley, the New Lobbying Monster*, NEW YORKER (Oct. 7, 2024), https://www.newyorker.com/magazine/2024/10/14/silicon-valley-the-new-lobbying-monster [https://perma.cc/D6Z4-6E3H].

[241] Coffee Jr., *supra* note 240, at 1020–37.

[242] Hayden Field, *AI Lobbying Spikes 185% as Calls for Regulation Surge*, CNBC (Feb. 2, 2024), https://www.cnbc.com/2024/02/02/ai-lobbying-spikes-nearly-200percent-as-calls-for-regulation-surge.html [https://perma.cc/863D-XPT4].

[243] *Id.*

Modernization Act of 2018.[244] Such laws provide presidential authority to intervene when foreign adversaries have undue influence and access to the data of Americans like through the use of popular social media app TikTok, which can serve as a platform for AI-driven misinformation.[245] Nevertheless, such unilateral executive actions could be subject to lengthy court challenges, thus stymying some of their impact.

Internationally, the political picture is not much better.[246] International relations and geopolitical considerations are critical impediments to swift, meaningful, and productive rulemaking and cooperation aimed at better regulating AI technologies in finance.[247] Key industry stakeholders in finance and technology aggressively lobby against regional and international rulemaking across a number of competing jurisdictions and regulatory bodies.[248] Some of this industry lobbying is rooted in not wanting any oversight, while some is understandably driven by a fear that crudely-crafted regulations may blunt the innovative progress and social benefits of AI.[249]

Furthermore, because of concerns about domestic markets, national security, and international competition, various nation-states often work to obstruct key reform efforts, placing short-term provincial political interests over long-term coordinated actions that could benefit the global financial system.[250]

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[244] *See* 21st Century Peace Through Strength Act, H.R. 8038, 118th Cong. div. D (2024); Foreign Investment Risk Review Modernization Act of 2018, Pub. L. No. 115-232, § 1702(c)(5), 132 Stat. 2174; Joseph R. Biden Jr., *Memorandum on Advancing the United States' Leadership in Artificial Intelligence; Harnessing Artificial Intelligence to Fulfill National Security Objectives; and Fostering the Safety, Security, and Trustworthiness of Artificial Intelligence*, WHITE HOUSE (Oct. 24, 2024), https://www.whitehouse.gov/briefing-room/presidential-actions/2024/10/24/memorandum-on-advancing-the-united-states-leadership-in-artificial-intelligence-harnessing-artificial-intelligence-to-fulfill-national-security-objectives-and-fostering-the-safety-security [https://perma.cc/S9L8-3R3S]; David E. Sanger, *Biden Administration Outlines Government 'Guardrails' for A.I. Tools*, N.Y. TIMES (Oct. 24, 2024), https://www.nytimes.com/2024/10/24/us/politics/biden-government-guidelines-ai.html.

[245] Anupam Chander & Paul Schwartz, *The President's Authority Over Cross-Border Data Flows*, 172 U. PA. L. REV. 1989, 1991–95 (2024).

[246] *See* Kristen E. Eichensehr, *Cyberwar & International Law Step Zero*, 50 TEX. INT'L L.J. 355, 356 (2015) ("New technologies pose challenges for law and for international law in particular. For as cumbersome and slow as domestic law appears in many circumstances, developing international law is often even more difficult.").

[247] Chris Brummer, *How International Financial Law Works (and How It Doesn't)*, 99 GEO. L.J. 257, 259–70 (2011).

[248] *See* Awrey & Judge, *supra* note 12, at 2307–08 (discussing challenges to financial regulation across international borders); DOWNES, *supra* note 202, at 18 ("Party politics and the influence of lobbyists and other nongovernmental organizations often stalemate each other, keeping real reform at bay.").

[249] *See* Satariano & Kang, *supra* note 16 (reporting on "fears that too many rules may inadvertently limit the technology's benefits").

[250] *See* Awrey & Judge, *supra* note 12, at 2308 (discussing how financial firms operating internationally are "often subject to oversight by different regulators who do not necessarily coordinate their regulation or supervision").

For instance, in 2024, European regulators expressed serious reservations about American firms engaging in anticompetitive practices as a means to dominate the global market for AI technologies in the future.[251] These reservations further hindered the implementation of meaningful regulations and public policies on existing AI technologies, despite the clear need for new rules in this emerging field.[252] Additionally, nation-states might impede international regulatory efforts so as to help bolster their domestic AI companies and initiatives to challenge those led by American firms and the United States.

In sum, the regulation of AI in financial markets is heavily influenced by domestic politics and international geopolitics because of the global nature of modern finance and technology.[253] Political systems, particularly in democratic states, are often designed to move gradually and minimize disruptions to the prevailing power structures.[254] As such, the absence of agile and responsive political processes in the United States and overseas presents one of the critical challenges to mitigating the deleterious effects of AI in finance while harvesting its positive outgrowths.[255]

V. KEY RECOMMENDATIONS

The emergence of AI-driven market manipulation and financial misconduct in our economy and marketplace has had, and will continue to have, significant implications for investors, firms, executives, nation-states, and policymakers. Key nation-states, like the United States, China, and the European Union, each have different and diverging preferred approaches to regulating AI and related technologies.[256] While a broad consensus, domestically and internationally, on the large regulatory and legal questions about AI and finance remains elusive, there are pragmatic near-term steps—building upon existing legal tools and powers—that can be taken to address the very real threats faced by regulators, firms, and investors on a daily basis. In particular, this Article recommends enhancing enforcement incentives and penalties for regulators, encouraging

[251] Steven Overly & Laurens Cerulus, *AI Choice Should Not Be 'American or American,' EU Antitrust Chief Warns*, POLITICO (Feb. 1, 2024), https://www.politico.eu/article/ai-should-not-be-american-or-american-eu-antitrust-chief-warns [https://perma.cc/4BB5-S2F3].

[252] Océane Herrero & Gian Volpicelli, *France's AI Hopes Collide with French Love of Regulating Tech*, POLITICO (Dec. 6, 2023) https://www.politico.eu/article/thierry-breton-mistral-battle-france-ai-soul-tech-regulation [https://perma.cc/UVP5-JQD7].

[253] *See, e.g.*, Brummer, *supra* note 247, at 259–60.

[254] *See* DOWNES, *supra* note 202, at 18 ("The processes and institutions that create laws are slow and methodical, replete with redundant checks and balances.").

[255] *See* Fletcher & Le, *supra* note 25, at 292 ("AI technology is developing more quickly than lawmakers can respond, putting lawmakers and regulators at a disadvantage vis-à-vis the entities and activities they are supposed to oversee."); DOWNES, *supra* note 202, at 18.

[256] *See, e.g.*, Whyman, *supra* note 231 (describing the varying approaches by key international states on AI regulation).

long-term passive investments for investors, and committing to revitalize traditional regulatory tools for policymakers.

A. *Enhance Incentives and Penalties*

Given the various policy tools at their disposal, financial regulators should use enforcement incentives and penalties to encourage financial intermediaries to better and more expeditiously manage the risks associated with AI-driven financial misconduct.[257] While rulemaking and legislative processes are slow, markets and technology move fast.[258] In order for regulators to timely respond to dangers in the marketplace, regulation by enforcement via litigation or adjudication is sometimes necessary.[259]

In addressing AI-driven financial misconduct, regulators should use enforcement incentives and penalties to help encourage better risk management by firms, particularly financial intermediaries who make up the core of the financial system.[260] These intermediaries, including asset managers, brokerages, commercial banks, clearinghouses, hedge funds, investment banks, mutual funds, and stock exchanges, are on the frontlines of market activities and form the modern financial infrastructure.[261] As such, regulators like those at the SEC and the DOJ should impose enhanced penalties for financial misconduct related to AI while exercising leniency for firms and persons that demonstrate good-faith efforts in building systems to detect and prevent such misconduct. These measures aim to encourage key intermediaries to help police the

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[257] *See, e.g.*, Donald C. Langevoort, *The SEC as a Lawmaker: Choices About Investor Protection in the Face of Uncertainty*, 84 WASH. U. L. REV. 1591, 1619–22 (2006) (discussing various policy tools of the SEC).

[258] *See* Fletcher & Le, *supra* note 25, at 292; DOWNES, *supra* note 202, at 17 ("[T]echnology changes exponentially, but social, economic, and legal systems change incrementally.").

[259] *See* Chris Brummer, Yesha Yadav & David Zaring, *Regulation by Enforcement*, 96 S. CAL. L. REV. 1297, 1298 (2024) ("One visible response by regulators to what they view as persistent—or particularly fast-moving—market challenges has been a sharp turn toward litigation to introduce or test out novel legal theories and frameworks that could have been the product or subject of legislative or administrative rulemaking.").

[260] *See* Yadav, *supra* note 196, at 1090–94 (arguing that intermediaries like stock exchanges should play a larger role in financial regulation and investor protection); Lin, *supra* note 185, at 661 (discussing the intermediated nature of modern finance); Gary Gorton & Andrew Winton, *Financial Intermediation*, *in* 1A HANDBOOK OF THE ECONOMICS OF FINANCE 432, 433 (George M. Constantinides et al. eds., 2003) ("Financial intermediation is a pervasive feature of all of the world's economies.").

[261] *See* Kathryn Judge, *Intermediary Influence*, 82 U. CHI. L. REV. 573, 614–24 (2015) (discussing the importance of financial intermediaries in the marketplace); BENTON E. GUP, BANKING AND FINANCIAL INSTITUTIONS: A GUIDE FOR DIRECTORS, INVESTORS, AND COUNTERPARTIES 23–24 (2011) (explaining the critical roles of key financial intermediaries).

marketplace and protect investors.[262] Additionally, to encourage better safeguards against unwitting dissemination of misinformation, regulators can be more aggressive in seeking the disgorgement of gains pilfered from AI-driven market manipulation. Because most securities-related cases settle, these enhanced incentives and penalties can be imposed as part of a negotiated settlement.[263]

To be clear, this recommended approach of using enforcement incentives and penalties contains both benefits and drawbacks. On the plus side, regulation by enforcement can swiftly and effectively address immediate concerns and infractions in rapidly changing financial markets, changing industry behavior expeditiously in the face of recalcitrant legislative impulses.[264] This approach is particularly pertinent in the realm of AI and finance where the underlying technology evolves at such high velocities, often outpacing the pace of formal regulatory development and implementation. Regulation by enforcement allows regulators to respond promptly to novel situations and infractions, maintaining market integrity and investor confidence through public, high-profile actions.[265] Thoughtful use of enforcement incentives and penalties provides a strong and flexible framework to deal with the unprecedented challenges posed by AI in finance, allowing regulators to adapt and experiment with enforcement and policymaking strategies according to the nuances of each new scenario.[266] At the same time, such public enforcement actions can incentivize firms to experiment creatively and learn to better safeguard their clients and themselves

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262 *See, e.g.*, Rory Van Loo, *Rise of the Digital Regulator*, 66 DUKE L.J. 1267, 1318–19 (2017) (discussing how private intermediaries can serve as key regulators in high-tech markets).

263 *See, e.g.*, Alexander I. Platt, *"Gatekeeping" in the Dark: SEC Control over Private Securities Litigation Revisited*, 72 ADMIN. L. REV. 27, 46 (2020) ("In reality, the vast majority of SEC enforcement actions are settled before trial, and as many as half of them are settled before they are even filed.").

264 *See* Gary Gensler, Chair, U.S. Sec. & Exch. Comm'n, Prepared Remarks at the Securities Enforcement Forum (Nov. 4, 2021), https://www.sec.gov/newsroom/speeches-statements/gensler-securities-enforcement-forum-20211104 [https://perma.cc/YV35-FP92] ("[H]igh-impact cases are important. They change behavior. They send a message to the rest of the market, to participants of various sizes, that certain misconduct will not be permitted. Some market participants may call this 'regulation by enforcement.' I just call it 'enforcement.'").

265 *See, e.g.*, Harvey L. Pitt & Karen L. Shapiro, *Securities Regulation by Enforcement: A Look Ahead at the Next Decade*, 7 YALE J. ON REG. 149, 155 (1990) ("Unlike many of its sister agencies, the SEC consistently has maintained a vigorous, highly-visible, and largely successful enforcement profile.").

266 *See, e.g.*, Zachary J. Gubler, *Experimental Rules*, 55 B.C. L. REV. 129, 136–39 (2014); Yair Listokin, *Learning Through Policy Variation*, 118 YALE L.J. 480, 483–84 (2008).

through more robust monitoring and compliance programs to avoid severe penalties.[267]

Admittedly, using enhanced enforcement incentives and penalties is not without drawbacks.[268] Regulation by enforcement can lead to a climate of uncertainty and unpredictability for firms.[269] It could lead to inconsistencies in the application of the law as decisions are made on a case-by-case basis without a standardized regulatory framework. This approach could also result in perceived or real unfairness and distrust where similar actions by different entities are treated differently, undermining the principle of equal treatment under the law.[270] Additionally, regulation by enforcement can initiate "a doom loop of legal challenges" between firms and regulators on matters of rulemaking and specific alleged infractions, ultimately exacerbating regulatory uncertainty and, potentially, systemic risk.[271]

In the context of AI and finance, this lack of clear, formalized regulations could lead to a fragmented landscape where businesses operate under a cloud of uncertainty regarding the legal implications of deploying AI capabilities in finance, thereby hampering the potential benefits these technologies might bring to firms and the greater public. Moreover, the 2024 U.S. Supreme Court ruling in *SEC v. Jarkesy*, which terminated the use of SEC administrative law tribunals to adjudicate securities fraud actions,[272] could make it more burdensome to bring enforcement actions.

Nevertheless, on balance, thoughtful and sensible use of enforcement incentives and penalties, accompanied by clear, publicly disclosed guidance, can serve as a powerful pragmatic tool against potential AI-driven financial misconduct. This may be particularly true following the 2024 Supreme Court ruling in *Loper Bright Enterprises v. Raimondo*, which overturned the longstanding *Chevron* doctrine, a principle of judicial deference to agencies

[267] *See* Susan Lorde Martin, *Compliance Officers: More Jobs, More Responsibility, More Liability*, 29 NOTRE DAME J.L. ETHICS & PUB. POL'Y 169, 177–82 (2015) (discussing how penalties and mitigating factors in connection with sentencing are key drivers of the growing compliance industry).

[268] *See, e.g.*, Brummer, Yadav & Zaring, *supra* note 259, at 1314–16 (surveying various criticisms of regulation by enforcement).

[269] *See* James J. Park, *The Competing Paradigms of Securities Regulation*, 57 DUKE L.J. 625, 663 (2007) ("[F]or the most part, the regulated prefer that regulators utilize rulemaking over principles-based enforcement actions . . . .").

[270] *See, e.g.*, COMM. ON CAP. MKTS. REGUL., INTERIM REPORT OF THE COMMITTEE ON CAPITAL MARKETS REGULATION 66 (2006), https://capmktsreg.org/wp-content/uploads/2022/11/Interim-Report-of-the-Committee-on-Capital-Markets-Regulation-1.pdf [https://perma.cc/XYU2-7CYQ] ("When new standards are introduced through specific enforcement actions and only later codified as explicit rules, confusion and distrust are likely to be the consequences.").

[271] *See* Brummer, Yadav & Zaring, *supra* note 259, at 1338; Benjamin P. Edwards, *Supreme Risk*, 74 FLA. L. REV. 543, 572–74 (2022).

[272] SEC v. Jarkesy, 603 U.S. 109, 140 (2024).

over ambiguous agency rules.[273] Thus, agency rulemaking and rule interpretations carry much less weight in a post-*Chevron* legal world.

Given new difficulties in making, interpreting, and enforcing agency rules, enhanced enforcement incentives and penalties—if applied properly—can work similarly to how the Federal Sentencing Guidelines and DOJ memos encourage companies to proactively build robust compliance and oversight systems within their own firms.[274] The Guidelines, as the name indicates, offer sentencing guidance for violations of federal law, including those related to corporate and financial misconduct, by establishing a scoring system and a Culpability Score.[275] The severity of the penalties corresponds with the Culpability Score; the higher the score, the more severe the penalty. Aggravating factors increase an offender's Culpability Score, while mitigating factors decrease it.[276] An "effective compliance and ethics program" is normally deemed a mitigating factor that decreases an offender's Culpability Score.[277] As such, it should be of little surprise that many firms invest significantly in creating robust compliance systems within their organizations, both as a means of *ex-ante* oversight and as a means of *ex-post* penalty mitigation.[278] In terms of AI-powered market misconduct, the careful use of enforcement incentives and penalties, accompanied by clear, publicly disclosed guidance, can similarly serve as a powerful tool to encourage firms to proactively and persistently respond to emerging threats in the marketplace.

B. *Encourage Long-Term Passive Investment Strategies*

One pragmatic policy and investment strategy in response to the accelerating integration of AI in finance and its associated risks is to encourage long-term passive investments for investors, particularly retail investors.[279] This recommendation is rooted in the inherently more volatile markets likely to arise

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[273] Loper Bright Enters. v. Raimondo, 603 U.S. 369, 412–13 (2024).

[274] *See generally* Sentencing Reform Act of 1984, Pub. L. No. 98-473, tit. II, ch. II, 98 Stat. 1987 (codified as amended in scattered sections of 28 U.S.C.); U.S. DEP'T OF JUST. CRIM. DIV., EVALUATION OF CORPORATE COMPLIANCE PROGRAMS (2024), https://www.justice.gov/criminal-fraud/page/file/937501/download [https://perma.cc/9U45-Y36X].

[275] U.S. SENT'G COMM'N, GUIDELINES MANUAL § 8C2.5 (2024), https://www.ussc.gov/sites/default/files/pdf/guidelines-manual/2024/GLMFull.pdf [https://perma.cc/9M6U-FUAA].

[276] *Id.*

[277] *See id.* § 8C2.5(f)(1).

[278] *See* James A. Fanto, *Advising Compliance in Financial Firms: A New Mission for the Legal Academy*, 8 BROOK. J. CORP., FIN. & COM. L. 1, 13–16 (2013) (discussing the growth of compliance functionaries within firms in response to changes in legal norms); Sean J. Griffith, *Corporate Governance in an Era of Compliance*, 57 WM. & MARY L. REV. 2075, 2077 (2016) ("Over the past decade, compliance has blossomed into a thriving industry, and the compliance department has emerged, in many firms, as the co-equal of the legal department.").

[279] *See* RICHARD A. FERRI, THE POWER OF PASSIVE INVESTING x–xviii (2011) (espousing the virtues of passive investing).

from AI-driven trading strategies, financial misconduct, and misinformation. To better sidestep the costly short-term and fleeting volatility of an AI-impacted market or security, investors are better served by focusing on long-term passive investments.[280]

AI algorithms, capable of processing vast amounts of data at unprecedented speeds, can trigger rapid market fluctuations, regardless of the integrity of the information.[281] When the information is accurate, this accelerated financial velocity leads to more efficient markets and greater price discovery.[282] However, when there is misinformation, this accelerated velocity can distort market prices in the very short term before the market corrects the distortion.[283] These swift and significant shifts are potentially profitable for high-frequency traders and large institutions with sophisticated AI capabilities, but they pose substantial risks for individual investors and the market's stability.[284]

Contemporary retail investors are particularly susceptible to the whims of these volatile markets because access to information and free trading apps are ubiquitous facts of modern financial life.[285] Within a few seconds and with a few taps on a phone, contemporary investors can easily execute a trade without paying a fee as if they were playing a game on their phone, only it is with their hard-earned money.[286] While convenient, such rapid trading is often deleterious to the portfolios and wealth of investors.[287] Such active trading historically has not been profitable for ordinary investors, who generally sell low and buy high.[288]

In contrast, passive investment strategies, such as investing in a low-fee S&P 500 index fund over a long time horizon, inherently promote

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[280] *See* Lin, *supra* note 40, at 1310 ("To best maximize their long-term returns in this turbulent, high-tech marketplace, ordinary investors should invest via low-fee index funds, exchange-traded funds, or mutual funds that track the broad marketplace.").

[281] *See supra* Part III.A.1.

[282] Brogaard, Hendershott & Riordan, *supra* note 136, at 2267.

[283] *See* Arcuri, Gandolfi & Russo, *supra* note 55, at 2; Marzo, *supra* note 60, at 171.

[284] PATTERSON, *supra* note 12, at 9–10.

[285] *See* Abraham J.B. Cable, *Regulating Democratized Investing*, 83 OHIO ST. L.J. 671, 674–77 (2022) (discussing the growth of retail investors using no-fee trading apps in capital markets).

[286] Tierney, *supra* note 73, at 356–67.

[287] *See, e.g.*, Brad M. Barber & Terrance Odean, *Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors*, 55 J. FIN. 773, 785–88 (2000); Andrea Frazzini & Owen A. Lamont, *Dumb Money: Mutual Fund Flows and the Cross-Section of Stock Returns*, 88 J. FIN. ECON. 299, 319 (2008) ("[I]ndividual investors have a striking ability to do the wrong thing."); Ronald C. Lease, Wilbur G. Lewellen & Gary G. Schlarbaum, *The Individual Investor: Attributes and Attitudes*, 29 J. FIN. 413, 429–32 (1974); Don A. Moore, Terri R. Kurtzberg, Craig R. Fox & Max H. Bazerman, *Positive Illusions and Forecasting Errors in Mutual Fund Investment Decisions*, 79 ORG. BEHAV. & HUM. DECISION PROCESSES 95, 110–12 (1999); Felix Salmon, *Stop Selling Bonds to Retail Investors*, 35 GEO. J. INT'L L. 837, 837–38 (2004).

[288] *See, e.g.*, Frazzini & Lamont, *supra* note 287, at 319.

diversification and reduce the risk exposure of individual investors to market anomalies and manipulative trading practices that may be amplified by AI algorithms.[289] Famed investor Warren Buffett has been a longtime advocate of this long-term passive investment strategy for most investors.[290] This is particularly sage advice in today's marketplace because by distributing investments across a wide range of assets, passive investors are less impacted by the volatility of individual stocks or sectors that may be disproportionately influenced by AI-driven trading activities, thereby offering a safer avenue for investors to grow their wealth.[291]

Furthermore, encouraging long-term passive investment strategies, especially for retail investors, arguably aligns with broader public policy objectives of promoting financial market stability and sustainable economic growth. Long-term-investment time horizons align investors' interests with the sustainable growth of companies and the economy, countering the myopic focus on fleeting, short-term speculative gains that could become even more prevalent in an AI-driven, high-tech financial marketplace.[292] Long-term investments could better reward companies and executives that plan and work toward generating long-term value for their investors. Collectively, such long-term thinking by many companies would ideally lead to a more sustainable economy.

In sum, long-term passive investment strategies are a pragmatic means to counteract the dangers of AI-driven market manipulation, misconduct, and misinformation. They can protect individual investors and also help foster a financial ecosystem that is better aligned with the broader economic goals of stability and sustainable growth.

C. *Revitalize Traditional Tools*

Greater revitalization of traditional financial regulation tools can serve as important weapons to safeguard financial markets and investors while policymakers debate large legislative and regulatory changes to better safeguard

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[289] BURTON G. MALKIEL, A RANDOM WALK DOWN WALL STREET: THE TIME-TESTED STRATEGY FOR SUCCESSFUL INVESTING 399–401 (2012).

[290] *See* Letter from Warren E. Buffett, Chairman of the Bd., Berkshire Hathaway Inc., to Berkshire Hathaway Shareholders 24 (Feb. 25, 2017), http://www.berkshirehathaway.com/letters/2016ltr.pdf [https://perma.cc/LC7Y-5PDT] ("Over the years, I've often been asked for investment advice, and in the process of answering I've learned a good deal about human behavior. My regular recommendation has been a low-cost S&P 500 index fund.").

[291] *Id.*

[292] For a discussion of the impact of financial "short-termism" on society and the economy, see, for example, Nadelle Grossman, *Turning a Short-Term Fling into a Long-Term Commitment: Board Duties in a New Era*, 43 U. MICH. J.L. REFORM 905, 906 (2010) ("[B]oard short-termism also seems to be due to some investors with short investment horizons who use activism to influence boards to make decisions that yield short-term returns despite the longer-term impairing effects those decisions might have on the corporate enterprise."), and Mark J. Roe, *Stock Market Short-Termism's Impact*, 167 U. PA. L. REV. 71, 74–75 (2018) (examining the wider impact of short-term investments and trading).

the marketplace and investors from misconduct using AI.[293] Part of this revitalization effort should smartly incorporate AI to help regulators detect misconduct, analyze legal problems, and design better practices more efficiently and effectively.[294] In particular, traditional regulatory tools—like public disclosures, human examinations, and stress tests—can be reinvented and reinvigorated in the near term to mitigate some of the harmful effects of AI on finance and society.[295] These time-tested regulatory mechanisms have significant potential as conceptual starting points for regulatory adaptation in the context of AI's unique challenges.[296]

1. *Reinvent Disclosures*

The transparency fostered by reimagining robust public disclosure requirements can be crucial in an AI-driven financial landscape.[297] Properly tailored enhanced disclosure norms and rules, like those in the securities market and banking industry, can ensure that firms provide clear information about the use, scope, impact, and limitations of AI technologies in their operations, even if such technologies can be black boxes at the coding level.[298] This informational transparency is not only essential for investor protection but also for maintaining market integrity and efficiency.[299] Furthermore, better disclosure practices can help regulators identify and safeguard risks on a more

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[293] *See, e.g.*, Satariano & Kang, *supra* note 16; Adam Satariano, *E.U. Agrees on Landmark Artificial Intelligence Rules*, N.Y. TIMES (Dec. 8, 2023), https://www.nytimes.com/2023/12/08/technology/eu-ai-act-regulation.html; Tugend, *supra* note 229.

[294] *See, e.g.*, Daniel Schwarcz & Jonathan H. Choi, *AI Tools for Lawyers: A Practical Guide*, 108 MINN. L. REV. HEADNOTES 1, 3–5 (2023) (discussing various ways in which AI can aid lawyers in their practice of law).

[295] *See, e.g.*, ERIK J. LARSON, THE MYTH OF ARTIFICIAL INTELLIGENCE: WHY COMPUTERS CAN'T THINK THE WAY WE DO 281 (2021) (arguing for "reinvesting in a culture of invention and human flourishing" as a means to confront the challenges posed by AI).

[296] *See, e.g.*, Kaminski, *supra* note 12, at 1380–86 (discussing how traditional regulatory tools can be modified to help regulate risks associated with AI).

[297] *See, e.g.*, Gina-Gail S. Fletcher, *Legitimate Yet Manipulative: The Conundrum of Open-Market Manipulation*, 68 DUKE L.J. 479, 542–43 (2018).

[298] *See, e.g.*, Robert P. Bartlett III, *Making Banks Transparent*, 65 VAND. L. REV. 293, 295–96 (2012) (arguing for improving disclosure practices and rules as a means for better financial regulation); Hu, *supra* note 154, at 1607–12 (recommending new disclosure norms that better account for new information technology); Steven L. Schwarcz, *Rethinking the Disclosure Paradigm in a World of Complexity*, U. ILL. L. REV. 1, 16–17 (2004).

[299] *See, e.g.*, Zohar Goshen & Gideon Parchomovsky, *The Essential Role of Securities Regulation*, 55 DUKE L.J. 711, 714–15 (2006); Ronald J. Gilson & Reinier H. Kraakman, *The Mechanisms of Market Efficiency*, 70 VA. L. REV. 549, 550–65 (1984); Eugene F. Fama, *Efficient Capital Markets: A Review of Theory and Empirical Work*, 25 J. FIN. 383, 404 (1970).

timely basis by counteracting managerial preferences for secrecy of economically sensitive and valuable information.[300]

If done well, greater transparency and better information will enable investors and regulators to better understand the underpinnings of AI-driven financial products and services, assess their reliability, and gauge the risks they could pose for the system as a whole.[301] Potential principles for driving disclosure improvements are changing the types of information that are disclosed and the methods of disclosure to better leverage technology.[302] For instance, enhanced disclosures could better account for the fact that AI technology can read and process a variety of information and data differently than disclosures intended only for human readers.[303] When reinvented and working well to better reflect new technology and microstructures of the marketplace, public disclosures can credibly and confidently continue to serve as the gold-standard source of market information.[304] Enhanced public disclosures should serve as a powerful clarion signal in a marketplace filled with a cacophony of noise.

2. *Reinvigorate Human Exams*

Regular examinations of financial institutions, a key pillar of traditional financial regulation, should be reinvented and reinvigorated to include assessments of AI governance, risk-management practices, and compliance with existing ethical standards.[305] Regulators, like those at the SEC's Division of Examination or the Federal Deposit Insurance Corporation, with the aid of new AI tools, can leverage these examinations to ensure that AI applications in finance align with broader systemic risk concerns, comply with existing legal frameworks, and follow policy principles such as fairness and nondiscrimination.[306] With more resource allocations, these exams can be

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[300] *See, e.g.*, Tom C.W. Lin, *Executive Trade Secrets*, 87 NOTRE DAME L. REV. 911, 941–44 (2012) (discussing the ways in which businesses use trade secrets and the law to shield information from public disclosure and scrutiny).

[301] *See, e.g.*, Uri Benoliel, *Have Disclosures Kept Up with the Big Data Revolution? An Empirical Test*, 63 B.C. L. REV. 1913, 1944–45 (2022).

[302] *See, e.g.*, *id.* (advocating for more machine-readable disclosures).

[303] *See, e.g.*, Sean Cao, Wei Jiang, Baozhong Yang & Alan L. Zhang, *How to Talk When a Machine Is Listening: Corporate Disclosure in the Age of AI*, 36 REV. FIN. STUD. 3603, 3604–07 (2023) (discussing ways to improve disclosures by leveraging AI).

[304] *See, e.g.*, Gerner-Beuerle, *supra* note 195, at 112–13 (arguing that "the disclosure paradigm of securities regulation of the pre-automation age should be enriched").

[305] *See, e.g.*, Julie Andersen Hill, *When Bank Examiners Get It Wrong: Financial Institution Appeals of Material Supervisory Determinations*, 92 WASH. U. L. REV. 1101, 1107 (2015).

[306] *See About the Division of Examinations*, U.S. SEC. & EXCH. COMM'N, https://www.sec.gov/exams/about [https://perma.cc/RU5G-P3LB] (June 29, 2024) (describing the work of the SEC's Division of Examinations). *See generally* DIV. OF RISK MGMT. SUPERVISION,

instrumental in better identifying potential risks arising from the use of AI, such as market manipulation, that may not be immediately apparent until key operators within firms are interviewed by a seasoned examiner. In fact, in 2024, the SEC's Division of Examinations disclosed that AI-related risks will be a priority area of focus going forward.[307]

Properly resourced, human exams and examiners are second to none and are still very much needed. Despite numerous advances in AI technology in law and compliance, as of late 2023, ChatGPT and other leading AI systems still could not properly review and analyze SEC securities filings with the acumen of human experts (though that will likely change in the near future with improved technology).[308] Human-to-human interactions, like those by bank and SEC examiners through "full-scope, on-site examination" at firms, are invaluable for unearthing risks and understanding bank operations.[309] Additionally, these exams can serve as secure forums for sharing sensitive information that then allow regulators important glimpses into emerging trends in the marketplace.[310]

In sum, reinvigorating human exams is a good way for regulators to help keep the "human in the loop," as technologists are fond of saying, as a means to better manage the deleterious effects of AI in the marketplace.[311]

3. *Reimagine Stress Tests*

Stress tests, traditionally used to evaluate a financial institution's capital resilience to adverse market conditions, can be recalibrated to include scenarios

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FED. DEPOSIT INS. CORP., RISK MANAGEMENT MANUAL OF EXAMINATION POLICIES (2024), https://www.fdic.gov/resources/supervision-and-examinations/examination-policies-manual/risk-management-manual-complete.pdf [https://perma.cc/3623-BC4X] (providing an overview of bank examinations); BRIAN CHRISTIAN, THE ALIGNMENT PROBLEM (2020) (explaining the challenges of aligning AI with human values).

307 DIV. OF EXAMINATIONS, U.S. SEC. & EXCH. COMM'N, 2024 EXAMINATION PRIORITIES 19 (2024), https://www.sec.gov/files/2024-exam-priorities.pdf [https://perma.cc/K64Y-BJAW] ("The Division remains focused on certain services, including automated investment tools, artificial intelligence, and trading algorithms or platforms, and the risks associated with the use of emerging technologies and alternative sources of data.").

308 Kif Leswing, *GPT and Other AI Models Can't Analyze an SEC Filing, Researchers Find*, CNBC (Dec. 19, 2023), https://www.cnbc.com/2023/12/19/gpt-and-other-ai-models-cant-analyze-an-sec-filing-researchers-find.html [https://perma.cc/CMU5-4AN5].

309 *See* 12 U.S.C. § 1820(d)(1). *But see* Hill, *supra* note 305, at 1161–65 (discussing various weaknesses of human bank examinations).

310 *See, e.g.*, Jose A. Lopez, *Using CAMELS Ratings to Monitor Bank Conditions*, FED. RES. BANK OF S.F. (June 11, 1999), https://fraser.stlouisfed.org/files/docs/historical/frbsf/frbsf_let/frbsf_let_19990611.pdf [https://perma.cc/9GL8-NGUY] ("[O]n-site bank exams seem to generate additional useful information beyond what is publicly available.").

311 MOLLICK, *supra* note 8, at 52.

that specifically address AI-related risks.[312] These might encompass testing for algorithmic biases, data integrity issues, or vulnerabilities to cyber threats, thereby ensuring that financial institutions and other critical firms are better prepared for a range of AI-related risks.[313] While no stress test can perfectly predict a calamitous event, a good stress test—perhaps enhanced with AI—can reveal vulnerabilities and other valuable information that firms can use to better plan ahead for certain risks in the same way that a good war game can prepare the military for actual battle without perfectly simulating war.[314] Such advanced planning makes thoughtfully designed stress tests a valuable tool to help mitigate the risks AI poses for our marketplace.[315]

Through stress testing, firms can develop technologies to bolster their cybersecurity to fill previously undisclosed weaknesses about AI-related risks, collect intrafirm intelligence, and create improved systems to simulate and monitor AI-powered financial misconduct more effectively.[316] Additionally, AI itself can be leveraged to run various stress test simulations in creative and cost-effective ways to help ensure the stability and integrity of firms and the marketplace writ large.[317]

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[312] *See* ALLEN, *supra* note 12, at 152 (discussing the use of stress tests as part of financial regulation); Lin, *supra* note 213, at 1431 ("Policymakers should design advanced technological stress tests to assess the information technology infrastructure of systemically important private and public financial institutions and agencies.").

[313] For a general discussion of stress testing, see Robert Weber, *A Theory for Deliberation-Oriented Stress Testing Regulation*, 98 MINN. L. REV. 2236, 2238–39 (2014), and Mehrsa Baradaran, *Regulation by Hypothetical*, 67 VAND. L. REV. 1247, 1283–84 (2014).

[314] *See, e.g.*, Baradaran, *supra* note 313, at 1319 ("The military has used war games for many years, both as a test of the military's responsiveness to crises and as a way to devise military strategies.").

[315] *See, e.g.*, Policy Statement on the Scenario Design Framework for Stress Testing, 12 C.F.R. pt. 252 app. A(1)(e) (2017) ("[Stress testing is] a valuable supervisory tool that provide[s] a forward-looking assessment of large financial companies' capital adequacy under hypothetical economic and financial market conditions.").

[316] *See, e.g.*, Nektaria Kaloudi & Jingyue Li, *AST-SafeSec: Adaptive Stress Testing for Safety and Security Co-Analysis of Cyber-Physical Systems*, 18 IEEE TRANSACTIONS ON INFO. FORENSICS & SEC. 5567, 5577–78 (2023), https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10231138 [https://perma.cc/WGW2-DXQF]; James A. "Sandy" Winnefeld Jr., Christopher Kirchhoff & David M. Upton, *Cybersecurity's Human Factor: Lessons from the Pentagon*, HARV. BUS. REV., Sept. 2015, at 86, 88, https://hbr.org/2015/09/cybersecuritys-human-factor-lessons-from-the-pentagon.

[317] *See, e.g.*, Mark Koren, Ahmed Nassar & Mykel J. Kochenderfer, *Finding Failures in High-Fidelity Simulation Using Adaptive Stress Testing and the Backward Algorithm*, 2021 IEEE/RSJ INT'L CONF. ON INTELLIGENT ROBOTS & SYS. 5944, 5949 (2021), https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9636072 [https://perma.cc/QDZ8-9QHK] (discussing the use of artificial research to find vulnerabilities in autonomous vehicle systems); Ritchie Lee et al., *Adaptive Stress Testing: Finding Likely Failure Events with Reinforcement Learning*, 69 J. A.I. RSCH. 1165, 1198 (2020).

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Reinvigorating and reinventing longstanding regulatory tools, like the ones discussed above, to address the nuances of AI in finance in addition to safeguarding the marketplace also has the added benefit of enhancing regulatory technical expertise and critical regulatory technological capabilities.[318] For instance, with more expertise and technology, regulators can better leverage AI machine learning to help with monitoring and supervision to generate faster, improved results at lower costs.[319] Such reinvestment and reinvention will also help attract talented individuals and newer systems to agencies like the SEC. To properly do their jobs in a financial marketplace driven by AI, regulators must be equipped with the necessary skills and tools to understand and evaluate AI technologies effectively.

In sum, through thoughtful revitalization of traditional regulatory tools like mandated disclosures, human exams, and stress tests, policymakers can create a robust framework that effectively oversees the growth of AI in finance, ensuring that it contributes to the sector's innovation while safeguarding against its potential risks to society.[320] To be sure, designing the detailed means to reinvigorate these traditional regulatory tools within the confines of pre-existing frameworks, budget allocations, and rapidly changing technology will not be easy, but it is necessary given the significant challenges of rulemaking and legislation.[321] These conceptual proposals offer practical starting points for this much-needed endeavor.

VI. CONCLUSION

Addressing critical issues arising at the nexus of AI, misinformation, market manipulation, and financial misconduct will be one of the most consequential and demanding challenges for public policymakers, financial regulators, business leaders, technologists, and many investors in the coming years.[322] These issues are forged in a complex, highly dynamic cauldron of money,

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[318] *See, e.g.*, Hilary J. Allen, *Resurrecting the OFR*, 47 J. CORP. L. 1, 21 (2021) (arguing for the development of more technological expertise by financial regulators as means to safeguard market stability); ORLY LOBEL, THE EQUALITY MACHINE 10 (2022) (discussing how regulation and public policy "can incentivize, leverage, and oversee technology" simultaneously).

[319] *See, e.g.*, Cary Coglianese & Alicia Lai, *Algorithm vs. Algorithm*, 72 DUKE L.J. 1281, 1281 (2022) ("Digital algorithms, such as machine learning, can improve governmental performance by facilitating outcomes that are more accurate, timely, and consistent.").

[320] *See, e.g.*, Mihailis E. Diamantis, *Vicarious Liability for AI*, 99 IND. L.J. 317, 334 (2023) (arguing for building on existing law as a practical means to regulate AI).

[321] *See supra* Part IV.A.

[322] *See* Fletcher & Le, *supra* note 25, at 315 ("Implementing a regulatory framework for AI in the financial markets is a difficult and wide-ranging feat. AI is complex and ever-changing . . . .").

technology, power, and politics that fuels our public and private institutions. How we address these issues and answer these questions will have profound ripples beyond law and finance.[323] While a perfectly detailed playbook for constructing precise solutions remains elusive, a workable guiding blueprint is within our reach.

This Article provides an original study to better understand, anticipate, and address the perilous threats to the stability and integrity of our marketplace posed by AI-driven misinformation and misconduct. It reveals structural vulnerabilities in our financial system exposed by AI, identifies regulatory challenges to governing AI in finance, and recommends workable next steps to better safeguard our markets and protect investors in the near term while larger political debates about how best to respond are unfolding around the world. In particular, it argues for enhancing incentives and penalties for AI-related actions, encouraging long-term passive investments, and reinvigorating traditional tools of financial regulation to confront the new challenges posed by AI.

Throughout its examination, this Article is fully aware of the heavy demands of responding to a highly dynamic and rapidly evolving technology like AI. At the same time, it is also fully cognizant of the need for swift, sensible action against the imminent threats that imperil our markets, economy, and society. This awareness is ultimately rooted in a cautious hope that AI will enhance our collective capabilities to govern the marketplace to better harness its highest social promises while taming the savageries of our worst financial impulses.[324] To that end, this Article aspires to serve as an optimistic, thoughtful, and pragmatic blueprint for the hard work ahead—for better navigating the emerging crossroads of AI, misinformation, and market misconduct.

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[323] *See, e.g.*, Mihir A. Desai, *What the Finance Industry Tells Us About the Future of AI*, HARV. BUS. REV. (Aug. 9, 2023), https://hbr.org/2023/08/what-the-finance-industry-tells-us-about-the-future-of-ai.

[324] *See, e.g.*, BOSTROM, *supra* note 12, at 7–13 (discussing and forecasting various ways and models whereby AI could bolster human potential and capabilities).